Zero Inflated Count Regression Article

1 Background

1.1 Zero Inflated Count Regression

Zero inflated count regression is a modification of a generalized linear model following a count distribution, like Poisson or Negative Binomial. When our data has more counted zeros than the model would predict, this usually causes the model to be overdispersed, and the effects of terms in the model to be *overestimated*. To adjust for this, we treat the zeros as being predicted by a logistic regression model, while all the points greater than zero are predicted by the specified count regression model.

Where a typical negative binomial regression has the form:

$$log(\mu_{Y_i|x_i}) = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_{p-1} x_{(p-1)i}$$

where $\mu_{Y_i|x_i}$ is the mean of the Negative Binomial distribution from which the ith response is drawn.

A zero inflated negative binomial model predicts the zeros from a logistic model:

$$log\left(\frac{\pi_i}{1-\pi_i}\right) = \alpha_0 + \alpha_1 x_{1i} + ... + \alpha_{p-1} x_{(p-1)i}$$

where $log\left(\frac{\pi_i}{1-\pi_i}\right)$ is the log odds of a structural 0.

Also, two models don't need to have the same set of predictor x_i's.

Note: The same transformation can be applied to poisson models as well as models with an offset.

We usually care about interpreting the negative binomial model more than the logistic model, but what causes zeros can also be interesting.

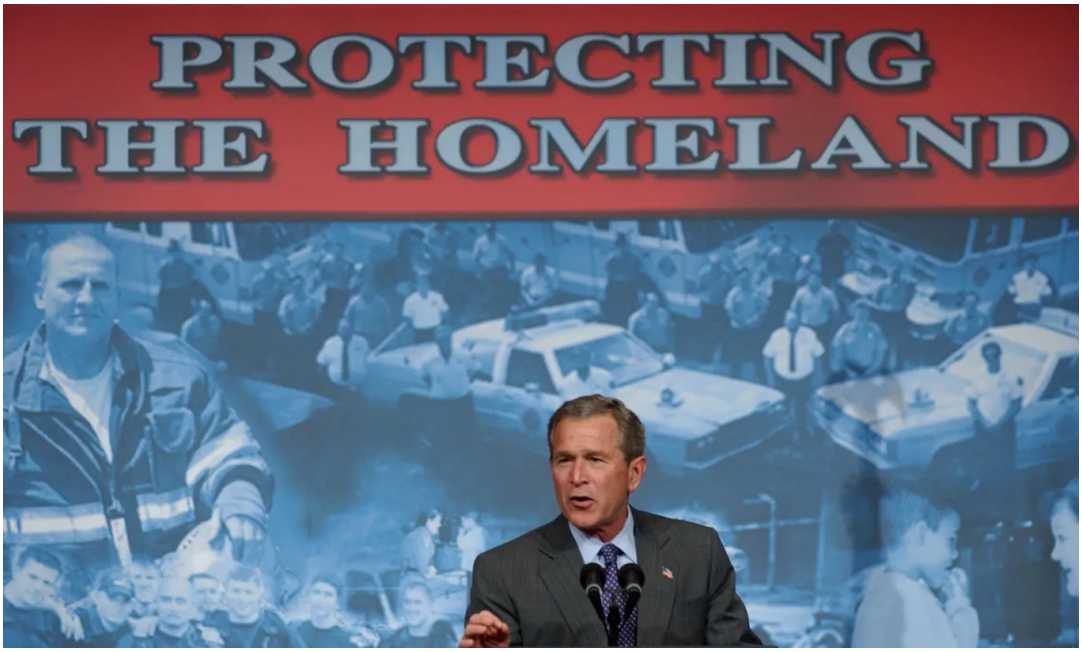
1.2 The Study

Figure 1: Nick Anderson



Surveillance, in any form, has long been controversial and politicized. Foucalt used the example of the panopticon to illustrate the way constant surveillance suffocates freedom and assures the maintenance of power (Foucault 2012). 25 Years ago, the war on terror in the United States served as the pretense for the passing of the Patriot Act, allowing the U.S. government greater access to personal digital information for the ostensible cause of anti-terrorism (King 2026).

Figure 2: Newsweek



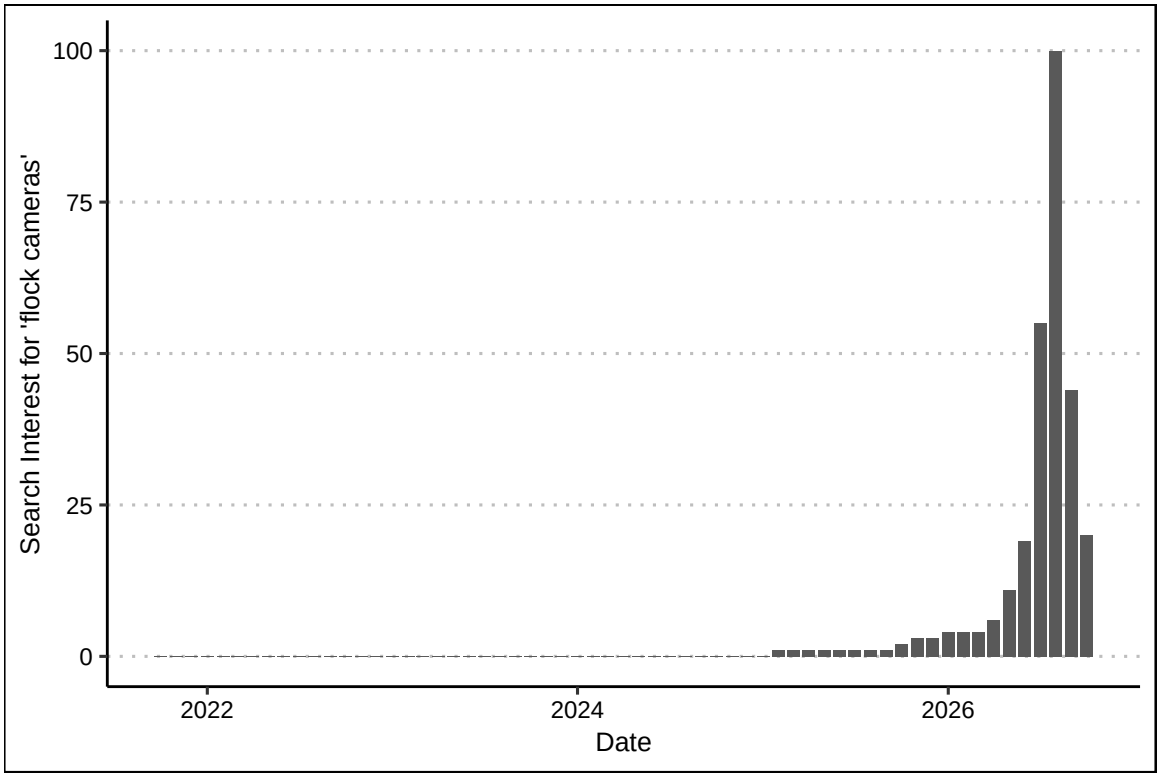
Where surveillance in previous decades was limited by an understanding that the guard in the panopticon tower could not be constantly watching every person, and a government official could not sift through every possible online document, the ability of large language models to scrape information from massive databases in real time has pushed the envelope for surveillance.

Figure 3: WBAL



Surveillance camera adoption has exploded in recent years, as algorithmic video detection models, like those used in cameras manufactured by Flock Safety, has led to data collection and surveillance on an unparalleled level (Brumbaugh 2026).

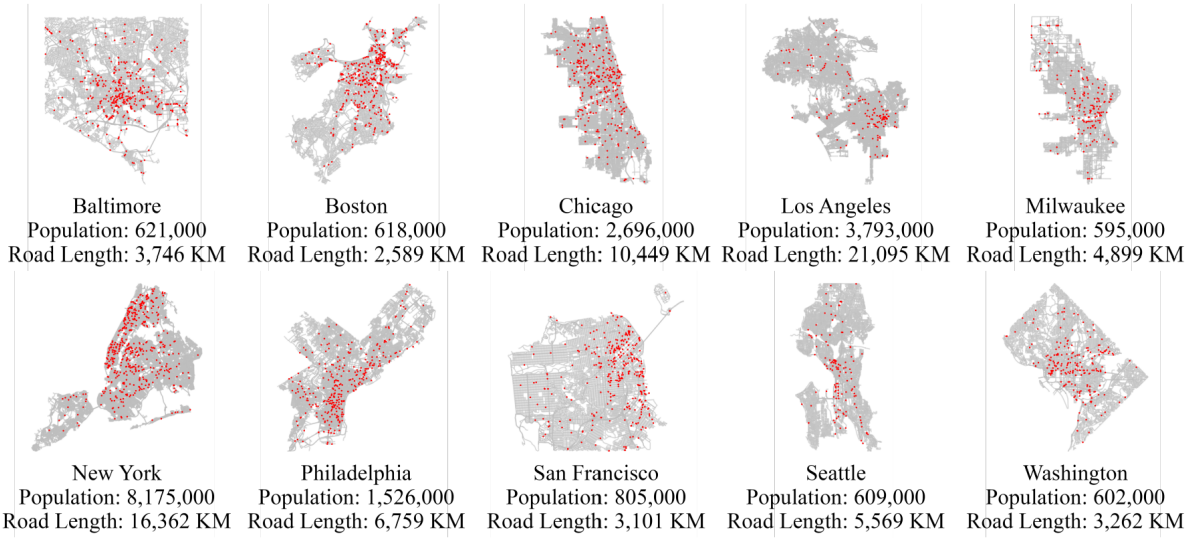
Figure 4: Google Trends of Flock Cameras



The article by Dahir et al. titled, "Surveillance camera prevalence and racial diversity in ten US cities", published in 2025, looks to see how the prevalence and density of surveillance cameras in major U.S. cities differs by demographic and socioeconomic characteristics of the areas they're in. They look to investigate if surveillance cameras are highest in homogeneous black neighbrhoods, as that would confirm the historical theory that black people often live in neighborhoods with characteristics resembling the carceral system. Another hypothesis says that surveillance cameras might be highest in more diverse neighborhoods where white residents might be more inclined to use the cameras for perceived safety and control. For example, white people who might be perceived as 'gentrifiers' in historically diverse neighborhoods might take pride in heterogeneous demographics while still feeling a sense of comfort with the added surveillance of a camera (Dahir et al. 2025).

To collect their count of surveillance cameras, the authors first trained a machine learning image detection model on confirmed camera locations the ten most dense populated cities in the United States:

Figure 5: Dahir et al.



After model training, they randomly selected 100,000 points within each city, went to each of those points with Google Street View, and captured a 360° photo of the surroundings. They they ran all those photos through the camera detection models, and had humans validate the existence of cameras in photos flagged by the model (those poor undergrads). The researchers estimate that around 60% of cameras in the photos were detected by the model, so the magnitude of the data is likely undercounted. The count of the cameras was then combined with publicly available demographic data from where each photo was taken to create a combined dataset.

Figure 6: Dahir et al.



1.3 Goal of Statistical Analysis

Because of the countable nature of the data, the researchers propose using a generalized linear model with a poisson distribution. Because of a large amount of images with no cameras taken, it makes sense to include the zeros in the model structurally with a logistic component.

I think their main analysis points are seeing how camera amounts change with regards to the white %, black%, and overall diversity of the areas, while accounting for other socioeconomic and strucutural confounding variables like income levels, crime rate, zoning, etc.

1.3.1 Formula Antics

I had a somewhat hard time determining the correct formula and terms for the regression equation. Below I'll illustrate several reasons why:

In the methods, the authors say:

> " We also expect excess zeros to be generated by a separate process from counts."

I didn't immediately know what this meant without looking at the code. After finding the equation I assume they used by looking in the code, I found this quote which is, in a choice I find puzzling, separate from the quote above:

> "The zero component accounts for the city and road length in all models."

Which, by confirming with their code, means the logistic element of their zero-inflated model is only regressed on the city with the road length as an offset. However, I don't understand why the other metrics in the model wouldn't also contribute to the detection of zero cameras, so I think it makes sense to leave them in both sides of the model.

They mention the inclusion of quadratic terms for the racial demographic variables (in the results but not the methods):

> "Given the parabolic relationships observed in the bivariate models, this model includes quadratic terms for both Black and white residents."

I take this to mean they adjusted the model after seeing a quadratic looking relationship in some of their effects in a "bivariate model". I can only assume this because there's no mention of a "bivariate model" in their methods, so the reader is left to wonder if that was the original hypothesis, a planned arm of their analysis, or something else. Still, the inclusion of the quadratic terms makes sense as the demographic variables are percentages and not truly continuous, however I'm not sure why they don't mention this explicitly in the methods. However, if these terms were only added because of a noticed trend in their data, this is basically 'HARKing', or "hypothesizing after the results are known," which is generally frowned upon in research (Taylor 2025).

Lastly, the authors mention the use of images taken as a weight component in the model:

> "We include … a weight for the image count."

This weights the data points with more images taken more heavily in the model. I think this highlights a potential issue with the data collection, and regardless of the choice of including or not including the weights, each has its own issues. If we don't include the images taken as a weight, a single image taken for one observation is weighted the same as an observation with over a hundred images taken; clearly one of those is more comprehensive. However, not all images are created equal, and images taken in close proximity to each other don't necessarily add importance to that observation. Because of this, I'll go forward with both options, seeing if the weights significantly alter the model.

With all that being said here's the regression equation I personally think makes sense:

Predictors (Poisson Counts and Logistic Zeros):

- % White (quadratic and linear)
- % Black (quadratic and linear)
- Overall Racial Diversity
- Crime Rate
- Population
- Median Household Income (in thousands)
- Home Vacancy Rate
- Median Home Value (in thousands)

- Neighborhood Designation (residential, public, commercial, industrial, roads, or mixed)
- City (Baltimore, Boston, Chicago, Los Angeles, Milwaukee, New York City, Philadelphia, San Francisco, Seattle, or Washington D.C.)

Offset:

- Road length (km)

Weights (potentially):

- log(images taken)

In *R*, our formula call looks like this:

```
formula <- cameras ~ white + I(white^2) + black + I(black^2) + diversity + crime +
    pop + income + vacancy + home_value + zone + city
```

2 Reproduction of the Study

2.1 Data Cleaning

I mainly renamed the variables to my liking, as well as scaling the continous metrics because that made the model behave better later on (not shown).

```
dat <- read_csv('cameras.csv')

dat <- dat |>
  rename(cameras = cam_count,
         images = image_count,
         white = pnhwht,
         black = pnhblk,
         diversity = entropy,
         crime = total_crime_rate,
         income = hinc,
         vacancy = pvac,
         home_value = mhmval,
         zone = modal_zone,
         road_length = road_length_km)

dat$diversity <- as.vector(scale(dat$diversity))
```

2.2 Data Summary

Let's see how the data is split up by city:

Our dataset is heavily influenced by New York City. We also see that most of the cities are relatively close in diversity, even with large differences in percentage of white and black populations. Lastly, there's not much we can get from the cameras because there's so many zeros.

Here's a rough correlation analysis between white%, black%, diversity% and cameras:

Hard to see any pattern in the demographic factors against the camera counts.

2.3 Model Assumptions

2.3.1 Zero Inflation

First let's see, how is zero inflated is the model compared to what we'd expect under a standard poisson regression:

We never even come close to simulating as many zeros as there are observed. This definitely warrants a zero inflated model.

Figure 7: Pairwise Summary of Data

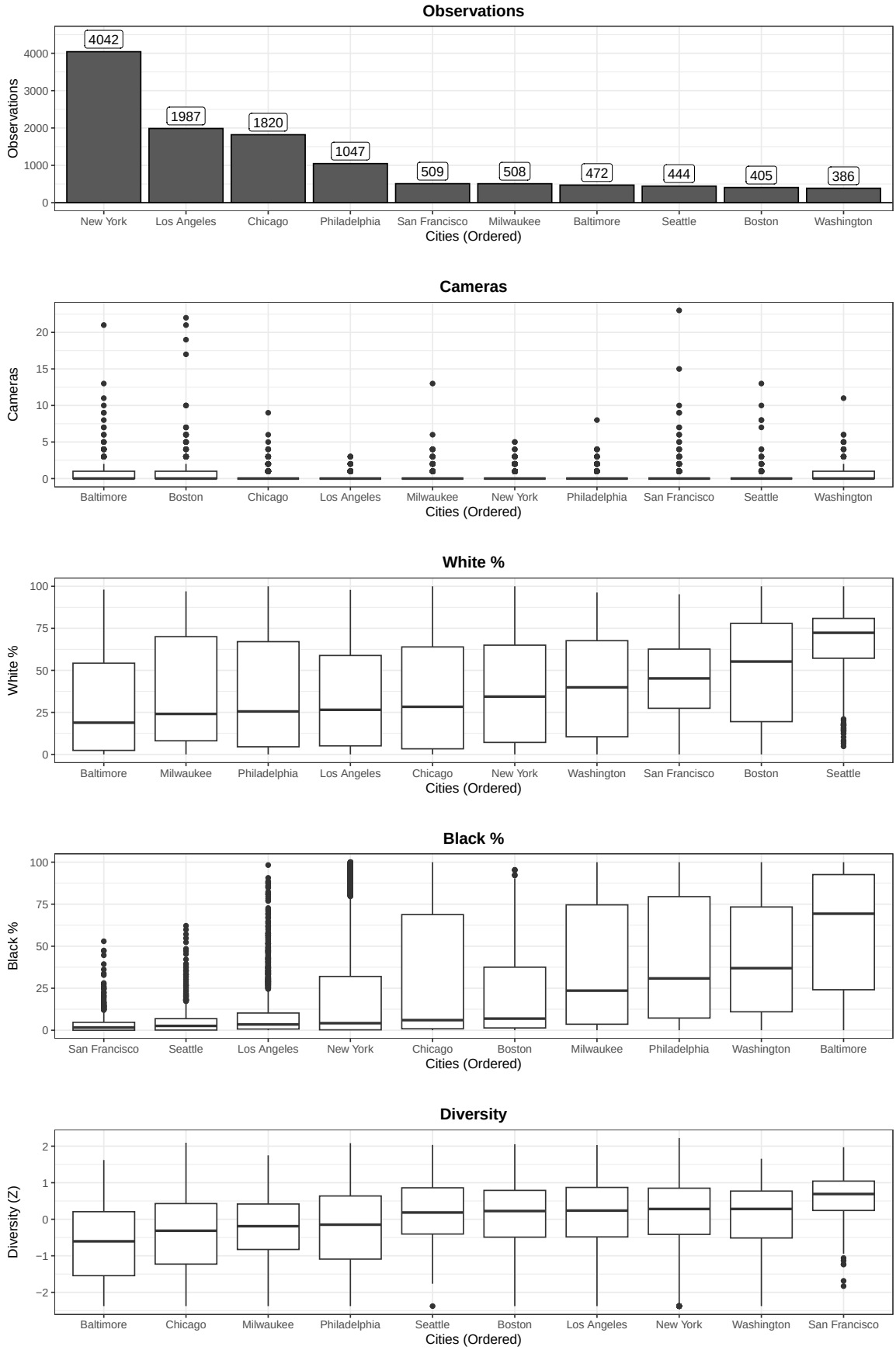


Figure 8: Correlation Analysis

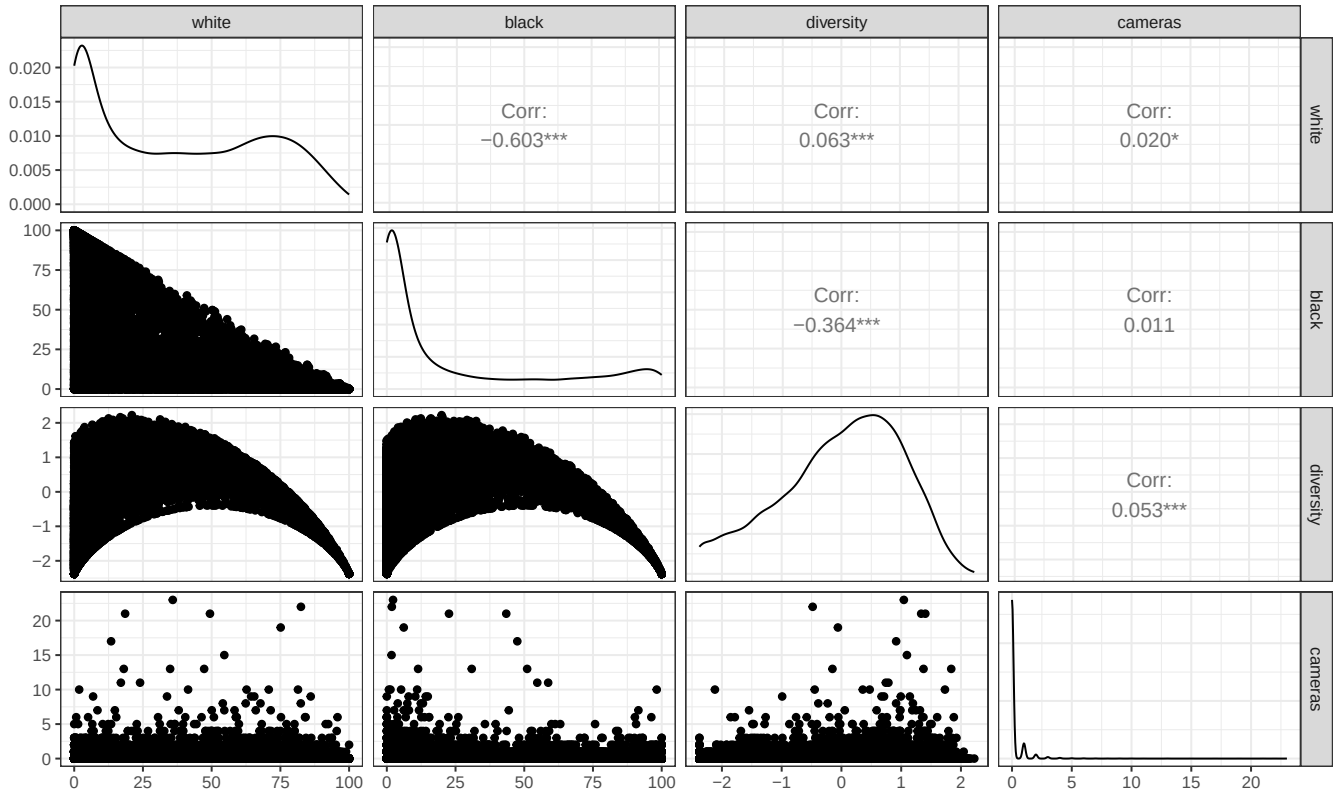
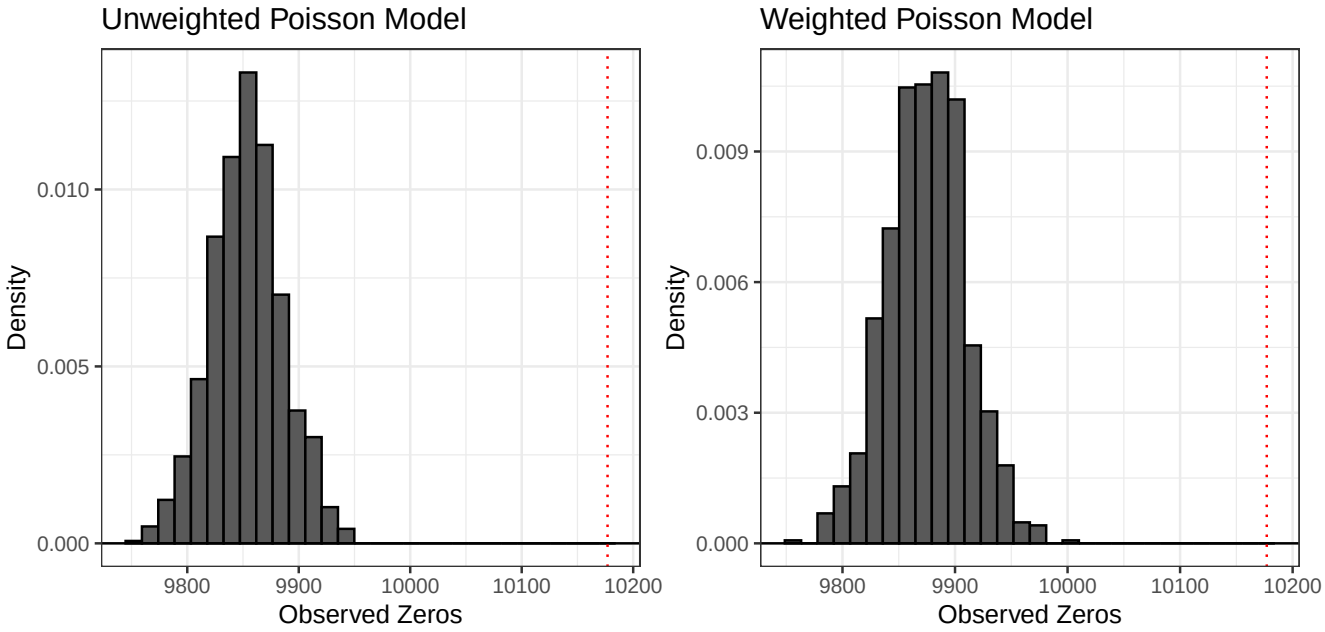


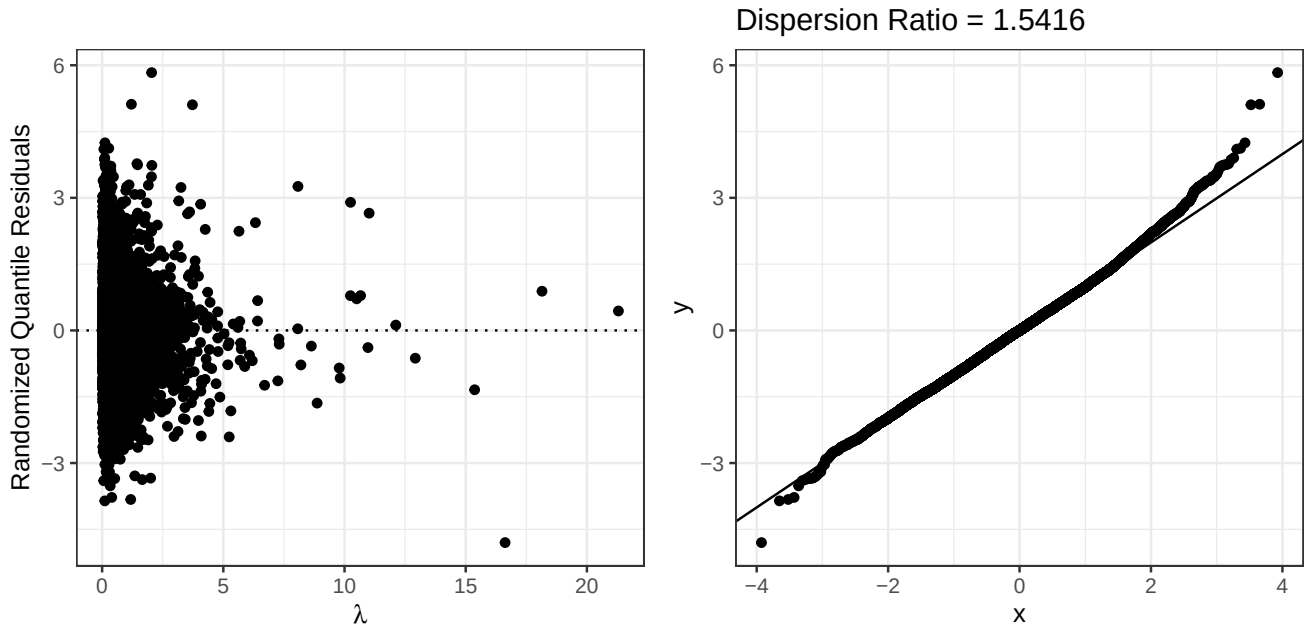
Figure 9: Zero Inflation Testing of the Standard Poisson Model



2.3.2 Dispersion

Here we'll test the randomized quantile residuals of the zero inflated model and the dispersion ratio. First, the unweighted model:

Figure 10: Randomized Quantile Residuals (Unweighted Model)



The randomized quantile residuals look pretty good, as there's a relatively even band of points. The dispersion ratio is a tad high but still not bad. There's a few outliers but that's to be expected with a very large sample size.

Next, the weighted model:

Adding the weights significantly increases the overdispersion of the model.

We can also check our predictions against the standardized Pearson residuals:

Looking more closely around 1, we see an increasing curved pattern as λ increases. This might suggest that a negative binomial might be better suited, but combined with the dispersion ratio and randomized quantile residuals, I think a poisson model is pretty well justified here.

The curved pattern seen before appears worse for the weighted model, but the points don't really show a clear trend, so this could be an artifact of random noise.

2.3.3 Multicollinearity

Because the design matrix is the same for both models, the VIF for both models is:

Table 1: Model VIF

Term	VIF	Lower CI	Upper CI
White %	4.772	4.643	4.883
(White %)^2	3.938	3.833	4.029
Black %	8.235	8.009	8.429
(Black %)^2	10.575	10.285	10.826
Diversity	7.794	7.581	7.978
Crime Rate	1.487	1.456	1.513
Population	1.425	1.397	1.449
Median Income	3.464	3.372	3.543

Table 1: Model VIF

Term	VIF	Lower CI	Upper CI
Vacancy Rate	1.368	1.342	1.390
Median Home Value	4.041	3.933	4.134
Zoning	23.407	22.770	23.972
City	100.364	97.664	102.821

To be expected, a lot of the variables have very high correlations with other variables, especially the demographic characteristics of each city. While I'll keep the city term in the model to compare with the paper's analysis, I do think it's probably not providing much necessary information that's not contained in other variables.

2.3.4 Sufficient Sample Size for Predictors

Ideally we should have at least 10 events counted per predictor:

$$\sum_{i=1}^{n=11,620} \lambda_i = 2524 \geq 10p = 10(25) = 250 \quad \checkmark$$

2.4 Model Fit

We can obtain an overall model fit of both models. First, the unweighted model:

Table 2: Model Fit

AIC	AICc	BIC	R2	R2_adjusted	RMSE	Sigma	Score_log	Score_spherical
10470.73	10471.17	10838.75	0.186	0.184	0.67	0.671	-0.446	0.009

Our unweighted model explains 18.4% of the variance in surveillance cameras.

Now the weighted model:

Table 3: Model Fit

AIC	AICc	BIC	R2	R2_adjusted	RMSE	Sigma	Score_log	Score_spherical
44517.89	44518.33	44885.91	0.062	0.06	1.522	3.598	-0.447	0.009

Our weighted model explains 6% of the variance in surveillance cameras.

Remember that some of the model fit statistics are non-comparable because they're calculated differently for weighted and non-weighted models.

We can directly compare the model fits with the cross-validated root mean square error:

Table 4: Cross Validated RMSEs of Both Models

Fold	unweighted	weighted
1	0.700	0.700
2	0.639	0.644
3	0.716	0.718
4	0.745	0.740
5	0.668	0.664
6	0.531	0.533

Table 4: Cross Validated RMSEs of Both Models

Fold	unweighted	weighted
7	0.965	0.990
8	0.672	0.673
9	0.739	0.733
10	0.687	0.689
Mean	0.706	0.708

This tells us, that by their ability to predict, the models are almost identical.

2.5 Model Inference

Now to the heart of the analysis:

- How many surveillance cameras are in highly white or highly black neighborhoods?
- How many surveilance cameras are in more diverse neighborhoods?
- Are there other confounding factors that affect surveillance cameras?

Spoiler: The effect of certain factors is dependent on the weighting of the model. As annoying as this is, it means I have to show a unweighted and weighted version of each analysis.

A general summary our unweighted model gives us:

Table 5: Unweighted Logistic Model Summary

	Estimate	SE	Z	p	significance	Percent Change
Intercept	-0.743	0.344	-2.157	0.031	*	
White %	0.183	0.134	1.362	0.173		20.1
(White %)^2	0.034	0.138	0.249	0.803		3.5
Black %	0.288	0.163	1.760	0.078	.	33.3
(Black %)^2	0.120	0.131	0.914	0.361		12.7
Diversity	0.146	0.160	0.911	0.362		15.7
Crime Rate	-1.798	0.216	-8.329	<0.001	***	-83.4
Population	-0.320	0.052	-6.123	<0.001	***	-27.4
Median Income	0.290	0.086	3.390	<0.001	***	33.7
Vacancy Rate	-0.167	0.067	-2.491	0.013	*	-15.3
Median Home Value	-0.450	0.114	-3.933	<0.001	***	-36.2
Zone = Industrial	-0.809	0.487	-1.662	0.096	.	-55.5
Zone = Mixed	-0.029	0.348	-0.082	0.934		-2.8
Zone = Public	0.389	0.346	1.124	0.261		47.5
Zone = Residential	0.438	0.215	2.040	0.041	*	54.9
Zone = Roads	-0.256	0.516	-0.495	0.621		-22.6
City = Boston	0.751	0.254	2.955	0.003	**	111.9
City = Chicago	0.006	0.302	0.019	0.985		0.6
City = Los Angeles	1.313	0.414	3.173	0.002	**	271.7
City = Milwaukee	-0.025	0.265	-0.095	0.924		-2.5
City = New York	1.014	0.295	3.437	<0.001	***	175.6
City = Philadelphia	-0.024	0.250	-0.097	0.922		-2.4
City = San Francisco	1.109	0.396	2.802	0.005	**	203.1
City = Seattle	1.332	0.306	4.346	<0.001	***	278.7
City = Washington	-0.205	0.476	-0.432	0.666		-18.6

where the percent change = $(e^{\text{estimate}} - 1) \times 100$.

Based on the summary of the unweighted logistic model, there's a:

Figure 11: Randomized Quantile Residuals (Weighted Model)

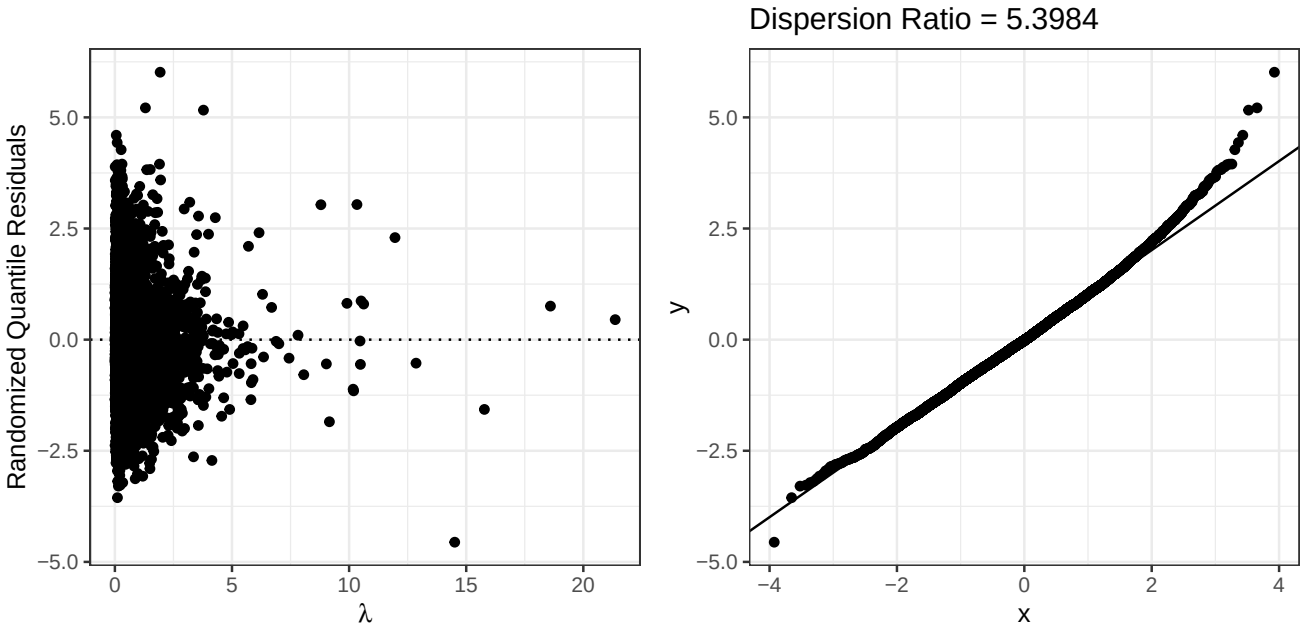


Figure 12: Predictions Against Pearson Residuals (Unweighted Model)

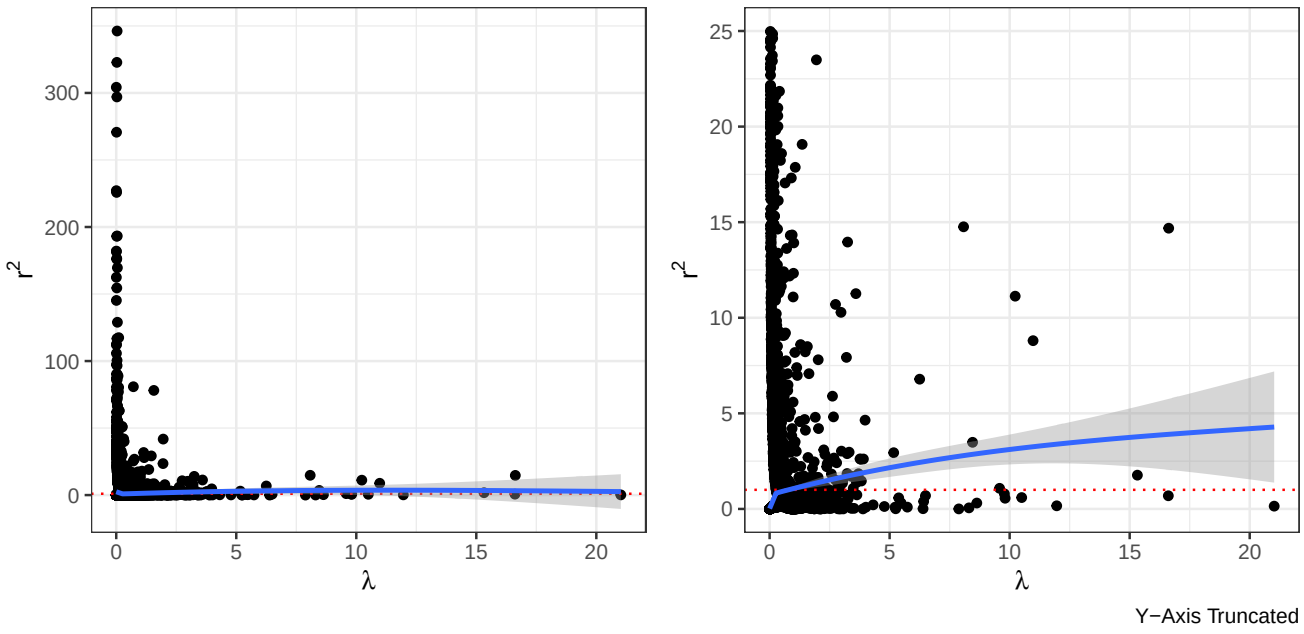
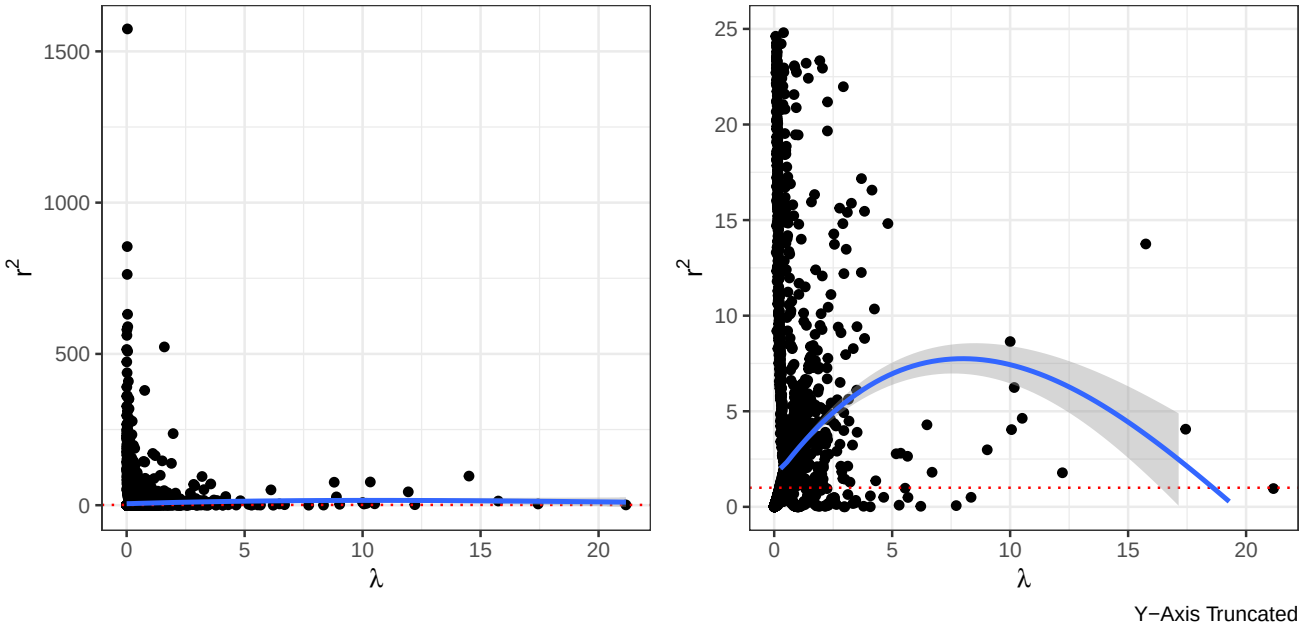


Figure 13: Predictions Against Pearson Residuals (Weighted Model)



- 15.7% greater chance of a surveillance camera as the diversity of the area increases by 1 standard deviation (non-statistically significant).
- 83.4% lesser chance of a surveillance camera as the crime rate of an area increases by 1 standard deviation ($Z = -8.329, p < 0.001$).
- 27.4% lesser chance of a surveillance camera as the population of an area increases by 1 standard deviation ($Z = -6.123, p < 0.001$).
- 33.7% greater chance of a surveillance camera as the median income of an area increases by 1 standard deviation ($Z = -3.390, p < 0.001$).
- 15.3% lesser chance of a surveillance camera as the vacancy rate of an area increases by 1 standard deviation ($Z = -2.491, p = 0.013$).
- 36.2% lesser chance of a surveillance camera as the median home value of an area increases by 1 standard deviation ($Z = -3.933, p < 0.001$).

We'll get into the other terms later on.

Table 6: Unweighted Count Model Summary

	Estimate	SE	Z	p	significance	Percent Change
Intercept	-0.985	0.146	-6.762	<0.001	***	
White %	0.005	0.065	0.074	0.941		0.5
(White %)^2	0.017	0.070	0.243	0.808		1.7
Black %	-0.090	0.081	-1.113	0.266		-8.6
(Black %)^2	-0.016	0.063	-0.247	0.805		-1.5
Diversity	0.107	0.076	1.406	0.16		11.2
Crime Rate	0.036	0.008	4.682	<0.001	***	3.6
Population	0.056	0.012	4.615	<0.001	***	5.8
Median Income	-0.086	0.041	-2.095	0.036	*	-8.2
Vacancy Rate	0.043	0.027	1.592	0.111		4.4
Median Home Value	-0.030	0.054	-0.555	0.579		-2.9
Zone = Industrial	-0.259	0.117	-2.216	0.027	*	-22.8
Zone = Mixed	0.144	0.106	1.353	0.176		15.4
Zone = Public	-0.426	0.145	-2.940	0.003	**	-34.7

Table 6: Unweighted Count Model Summary

	Estimate	SE	Z	p	significance	Percent Change
Zone = Residential	-0.495	0.087	-5.682	<0.001	***	-39
Zone = Roads	-0.509	0.316	-1.611	0.107		-39.9
City = Boston	0.128	0.112	1.146	0.252		13.7
City = Chicago	-1.942	0.124	-15.623	<0.001	***	-85.7
City = Los Angeles	-2.799	0.238	-11.744	<0.001	***	-93.9
City = Milwaukee	-1.246	0.127	-9.779	<0.001	***	-71.2
City = New York	-1.196	0.167	-7.177	<0.001	***	-69.8
City = Philadelphia	-1.197	0.112	-10.677	<0.001	***	-69.8
City = San Francisco	-0.632	0.187	-3.373	<0.001	***	-46.9
City = Seattle	-1.482	0.150	-9.868	<0.001	***	-77.3
City = Washington	-0.551	0.307	-1.793	0.073	.	-42.4

Based on the summary of the unweighted count model, we'd expect to see:

- 11.2% more surveillance cameras as the diversity of the area increases by 1 standard deviation (non-statistically significant).
- 3.6% more surveillance cameras as the crime rate of an area increases by 1 standard deviation ($Z = 4.682, p < 0.001$).
- 5.8% more surveillance cameras as the population of an area increases by 1 standard deviation ($Z = 4.615, p < 0.001$).
- 8.2% fewer surveillance cameras as the median income of an area increases by 1 standard deviation ($Z = -2.095, p = 0.036$).
- 4.4% more surveillance cameras as the vacancy rate of an area increases by 1 standard deviation (non-statistically significant).
- 2.9% fewer surveillance cameras as the median home value of an area increases by 1 standard deviation (non-statistically significant).

Note that the effect of the linear black % term is only marginally discernible, and the quadratic effect is non statistically discernible.

Table 7: Weighted Logistic Model Summary

	Estimate	SE	Z	p	significance	Percent Change
Intercept	-0.737	0.165	-4.473	<0.001	***	
White %	0.118	0.066	1.781	0.075	.	12.5
(White %)^2	0.046	0.066	0.702	0.483		4.8
Black %	0.341	0.079	4.302	<0.001	***	40.6
(Black %)^2	0.079	0.062	1.261	0.207		8.2
Diversity	0.126	0.077	1.633	0.102		13.5
Crime Rate	-2.012	0.106	-19.023	<0.001	***	-86.6
Population	-0.346	0.026	-13.438	<0.001	***	-29.3
Median Income	0.349	0.042	8.389	<0.001	***	41.7
Vacancy Rate	-0.129	0.031	-4.205	<0.001	***	-12.1
Median Home Value	-0.528	0.060	-8.740	<0.001	***	-41
Zone = Industrial	-1.152	0.264	-4.359	<0.001	***	-68.4
Zone = Mixed	-0.068	0.166	-0.408	0.683		-6.6
Zone = Public	0.248	0.166	1.499	0.134		28.2
Zone = Residential	0.320	0.103	3.096	0.002	**	37.7
Zone = Roads	-0.388	0.230	-1.682	0.093	.	-32.1
City = Boston	0.924	0.118	7.838	<0.001	***	152
City = Chicago	-0.035	0.149	-0.237	0.812		-3.5
City = Los Angeles	1.461	0.203	7.203	<0.001	***	331
City = Milwaukee	-0.070	0.120	-0.586	0.558		-6.8

Table 7: Weighted Logistic Model Summary

	Estimate	SE	Z	p	significance	Percent Change
City = New York	1.027	0.148	6.935	<0.001	***	179.3
City = Philadelphia	-0.055	0.120	-0.461	0.645		-5.4
City = San Francisco	1.209	0.192	6.285	<0.001	***	235.1
City = Seattle	1.484	0.141	10.532	<0.001	***	341.1
City = Washington	-0.192	0.212	-0.907	0.364		-17.5

Based on a summary of the weighted logistic model, there's a:

- 13.5% greater chance of a surveillance camera as the diversity of the area increases by 1 standard deviation (non-statistically significant).
- 86.6% lesser chance of a surveillance camera as the crime rate of an area increases by 1 standard deviation ($Z = -19.023, p < 0.001$).
- 29.3% lesser chance of a surveillance camera as the population of an area increases by 1 standard deviation ($Z = -13.438, p < 0.001$).
- 41.7% greater chance of a surveillance camera as the median income of an area increases by 1 standard deviation ($Z = 8.389, p < 0.001$).
- 12.1% lesser chance of a surveillance camera as the vacancy rate of an area increases by 1 standard deviation ($Z = -4.205, p = 0.013$).
- 41% lesser chance of a surveillance camera as the median home value of an area increases by 1 standard deviation ($Z = -8.389, p < 0.001$).

Table 8: Weighted Count Model Summary

	Estimate	SE	Z	p	significance	Percent Change
Intercept	-0.967	0.064	-15.074	<0.001	***	
White %	-0.041	0.029	-1.404	0.16		-4
(White %)^2	0.052	0.030	1.723	0.085	.	5.4
Black %	-0.092	0.035	-2.596	0.009	**	-8.8
(Black %)^2	-0.021	0.028	-0.764	0.445		-2.1
Diversity	0.127	0.034	3.778	<0.001	***	13.6
Crime Rate	0.038	0.003	11.802	<0.001	***	3.9
Population	0.058	0.005	11.711	<0.001	***	6
Median Income	-0.070	0.018	-3.962	<0.001	***	-6.8
Vacancy Rate	0.072	0.012	6.123	<0.001	***	7.5
Median Home Value	-0.051	0.025	-2.047	0.041	*	-5
Zone = Industrial	-0.321	0.050	-6.477	<0.001	***	-27.5
Zone = Mixed	0.090	0.046	1.947	0.052	.	9.4
Zone = Public	-0.472	0.063	-7.513	<0.001	***	-37.6
Zone = Residential	-0.569	0.038	-14.850	<0.001	***	-43.4
Zone = Roads	-0.557	0.137	-4.065	<0.001	***	-42.7
City = Boston	0.133	0.048	2.768	0.006	**	14.2
City = Chicago	-2.006	0.055	-36.370	<0.001	***	-86.5
City = Los Angeles	-2.831	0.110	-25.827	<0.001	***	-94.1
City = Milwaukee	-1.301	0.054	-24.206	<0.001	***	-72.8
City = New York	-1.347	0.078	-17.227	<0.001	***	-74
City = Philadelphia	-1.317	0.050	-26.170	<0.001	***	-73.2
City = San Francisco	-0.676	0.082	-8.248	<0.001	***	-49.1
City = Seattle	-1.469	0.063	-23.199	<0.001	***	-77
City = Washington	-0.560	0.134	-4.181	<0.001	***	-42.9

Based on the summary of the unweighted count model, we'd expect to see:

- 13.6% more surveillance cameras as the diversity of the area increases by 1 standard deviation ($Z = 3.778, p < 0.001$).
- 3.9% more surveillance cameras as the crime rate of an area increases by 1 standard deviation ($Z = 11.802, p < 0.001$).
- 6% more surveillance cameras as the population of an area increases by 1 standard deviation ($Z = 11.711, p < 0.001$).
- 6.8% fewer surveillance cameras as the median income of an area increases by 1 standard deviation ($Z = -3.962, p < 0.001$).
- 7.5% more surveillance cameras as the vacancy rate of an area increases by 1 standard deviation ($Z = 6.123, p < 0.001$).
- 5% fewer surveillance cameras as the median home value of an area increases by 1 standard deviation ($Z = -2.047, p = 0.041$).

Note that the quadratic effect of black % is not discernible on the model.

From now on, because we really care about the counts of the cameras, not as much the binary of no cameras versus cameras, we'll focus on the count regression factors going forward.

Because the other terms are either tangled in polynomial interactions or way too long to list out, we'll look at terms individually now:

2.6 Percentage of Black Residents

Table 9: Black % Estimated Marginal Trend (Unweighted)

Black % (Z) Trend	SE	Z	p	Percent Change	Lower CI	Upper CI
-0.016	0.005	-3.195	0.001	-1.567	-2.516	-0.609

Figure 14: Black % Estimated Marginal Trend (Unweighted)

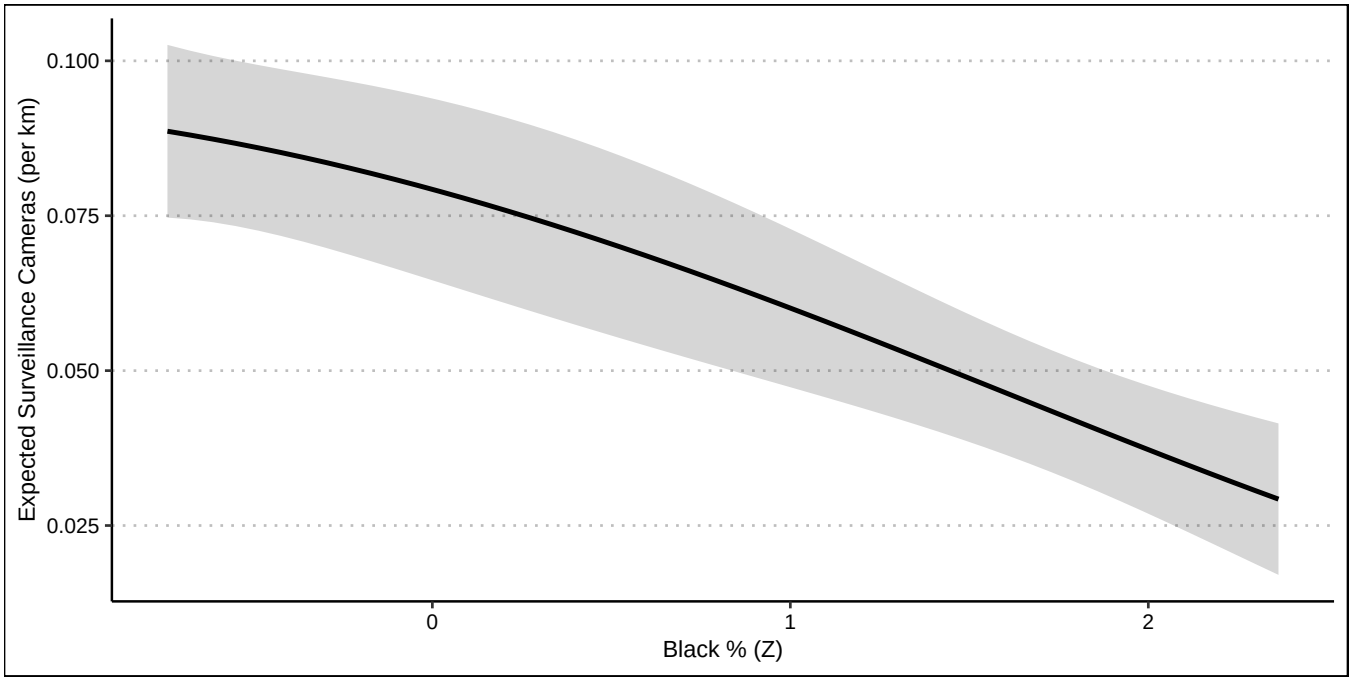
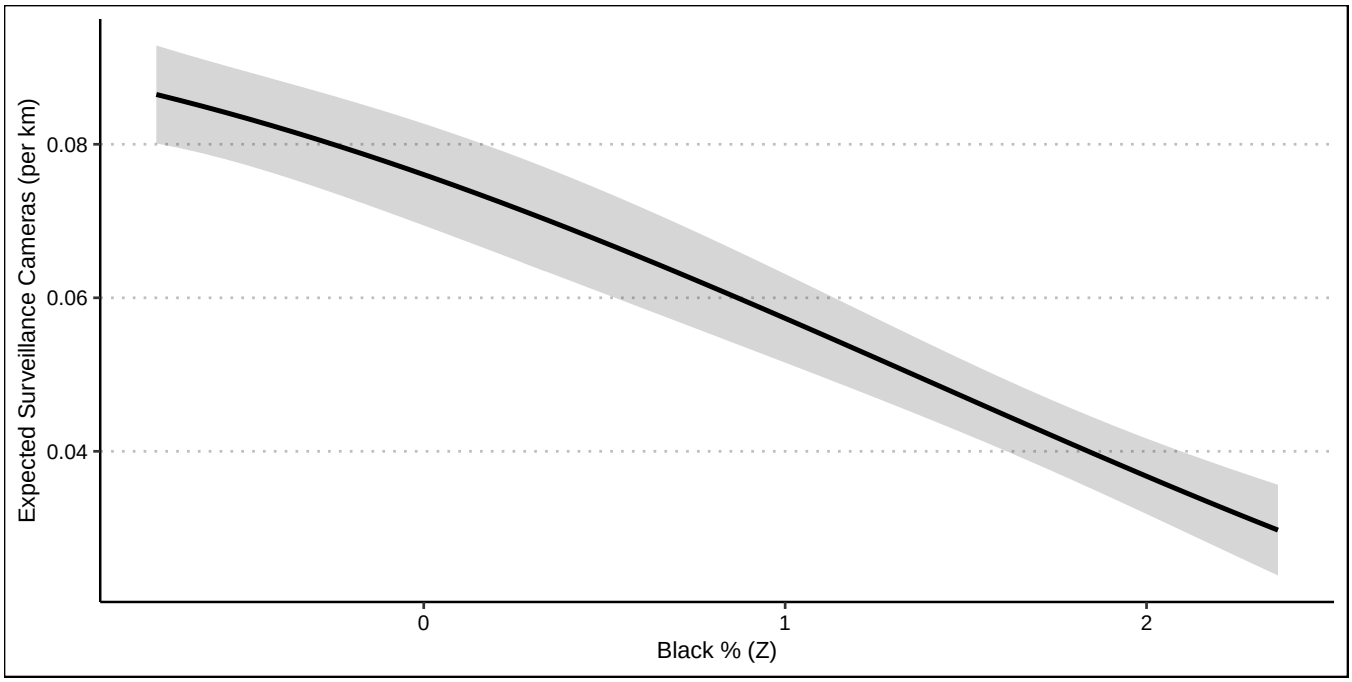


Table 10: Black % Estimated Marginal Trend (Weighted)

Black % (Z) Trend	SE	Z	p	Percent Change	Lower CI	Upper CI
-0.016	0.002	-7.271	<0.001	-1.631	-2.067	-1.194

Figure 15: Black % Estimated Marginal Trend (Weighted)



Notice that the trend of the factor is the same, it's just that the weighted models are have much tighter standard errors, and therefore greater appearing statistical significance. For the effect of the percentage of black residents in an area, the effect is still seen significantly across both models, but that won't always be the case as we'll see later.

2.7 Diversity of Residents

Table 11: Diversity Estimated Marginal Trend (Unweighted)

Diversity (Z) Trend	SE	Z	p	Percent Change	Lower CI	Upper CI
0.004	0.004	0.959	0.337	0.407	-0.422	1.243

Table 12: Diversity Estimated Marginal Trend (Weighted)

Diversity (Z) Trend	SE	Z	p	Percent Change	Lower CI	Upper CI
0.006	0.002	3.292	<0.001	0.62	0.25	0.99

In the case of the diversity of the residents, the weighted model is much more confident about the effect compared to the uncertainty in the unweighted model. This leaves us in a tricky spot, where your choice of how to view these results depends on how much you agree with the researchers decision to weight the model.

2.8 Crime Rate

Figure 16: Diversity Estimated Marginal Trend (Unweighted)

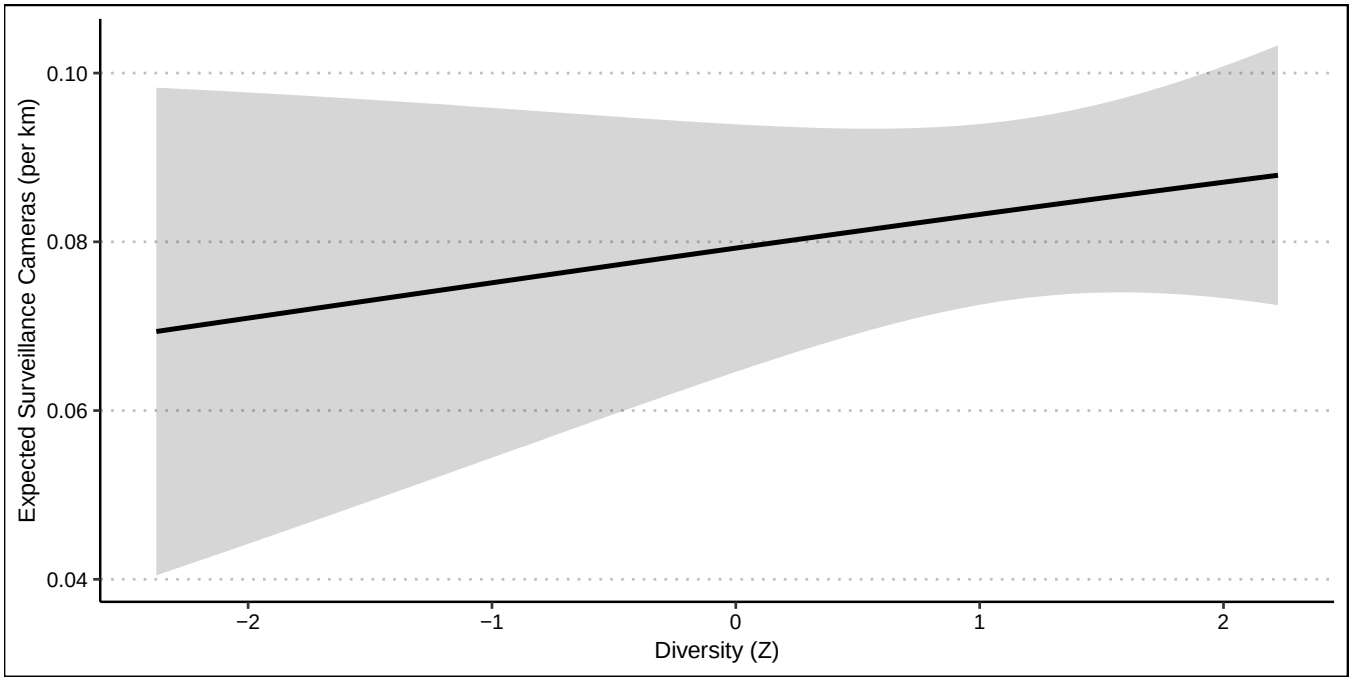


Figure 17: Diversity Estimated Marginal Trend (Weighted)

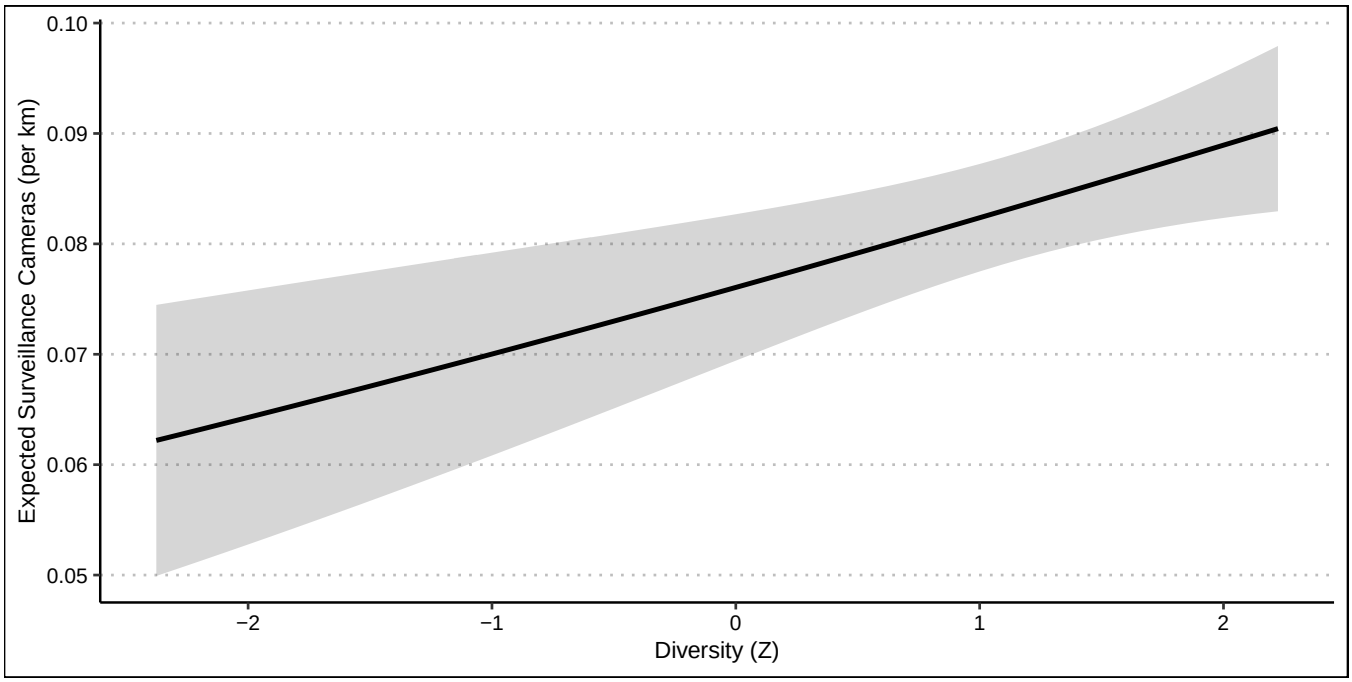


Table 13: Crime Rate Estimated Marginal Trend (Unweighted)

Crime Rate (Z) Trend	SE	Z	p	Percent Change	Lower CI	Upper CI
0.057	0.01	5.866	<0.001	5.836	3.849	7.86

Figure 18: Crime Rate Estimated Marginal Trend (Unweighted)

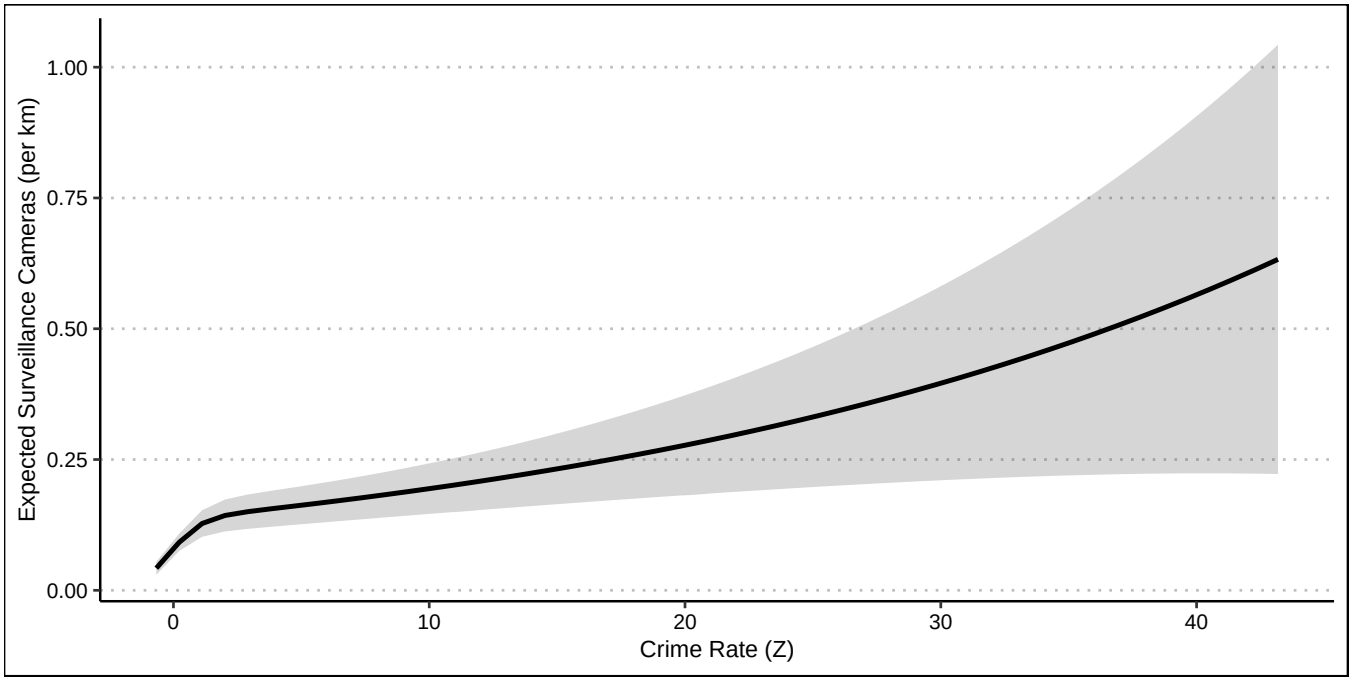


Table 14: Crime Rate Estimated Marginal Trend (Weighted)

Crime Rate (Z) Trend	SE	Z	p	Percent Change	Lower CI	Upper CI
0.058	0.004	13.273	<0.001	6.02	5.109	6.939

The effect of crime rate is pretty consistently positive on the amount of surveillance cameras. Because there are some crazy high outliers, both models are more uncertain towards the right tail of the data.

2.9 Population

Table 15: Population Estimated Marginal Trend (Unweighted)

Population (Z) Trend	SE	Z	p	Percent Change	Lower CI	Upper CI
0.014	0.002	6.469	<0.001	1.416	0.985	1.849

Table 16: Population Estimated Marginal Trend (Weighted)

Population (Z) Trend	SE	Z	p	Percent Change	Lower CI	Upper CI
0.014	0.001	14.287	<0.001	1.414	1.219	1.61

Both models are in agreement about the increasing number of surveillance cameras in areas with higher and higher populations.

Figure 19: Crime Rate Estimated Marginal Trend (Weighted)

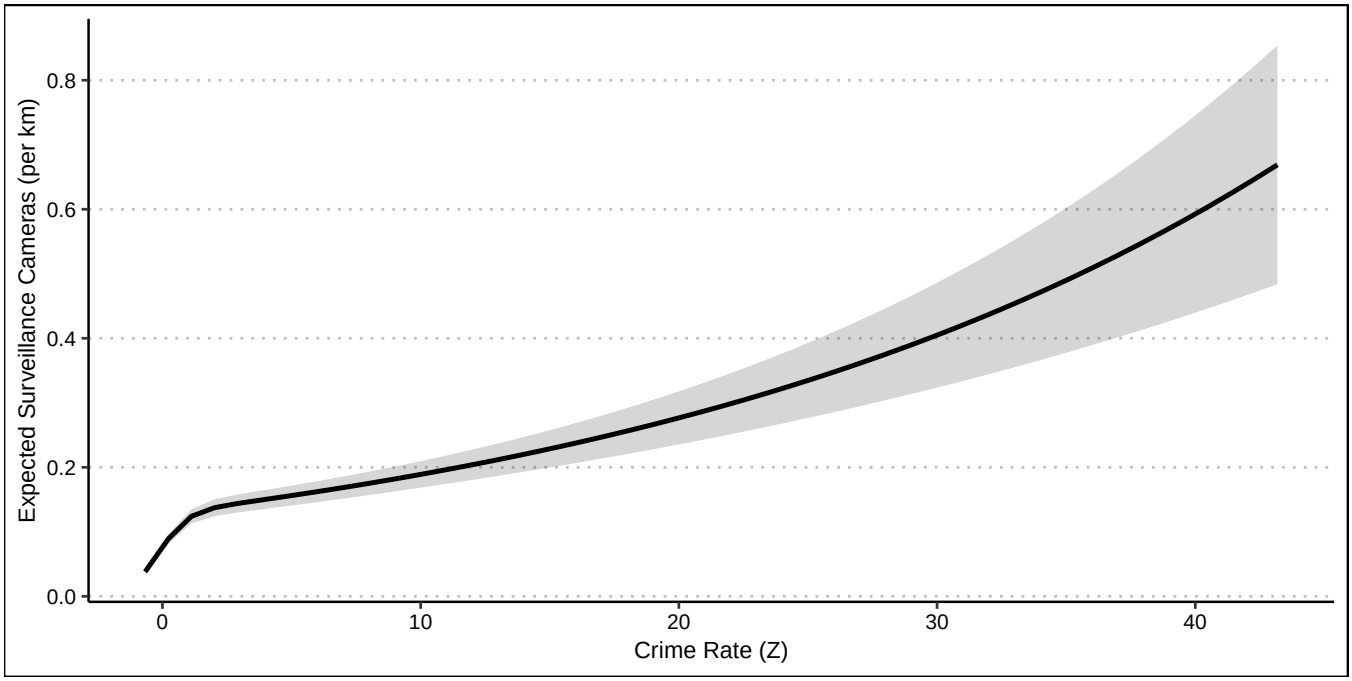


Figure 20: Population Estimated Marginal Trend (Unweighted)

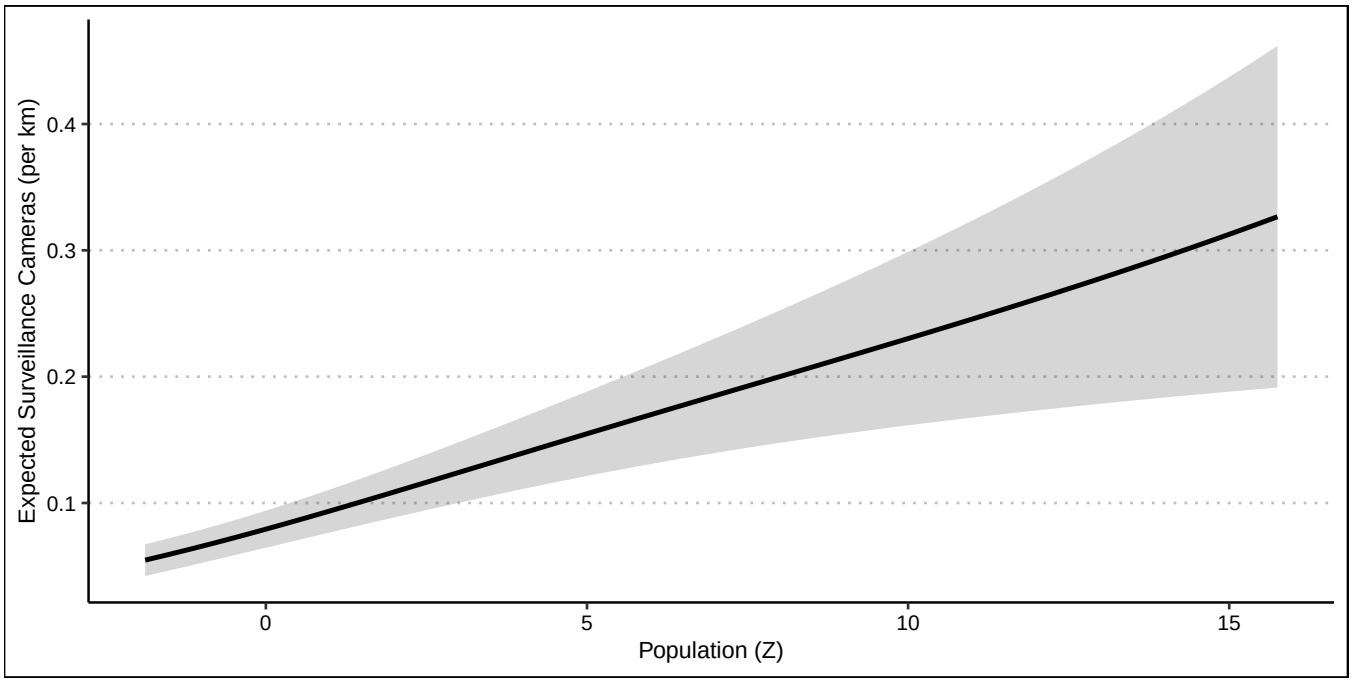
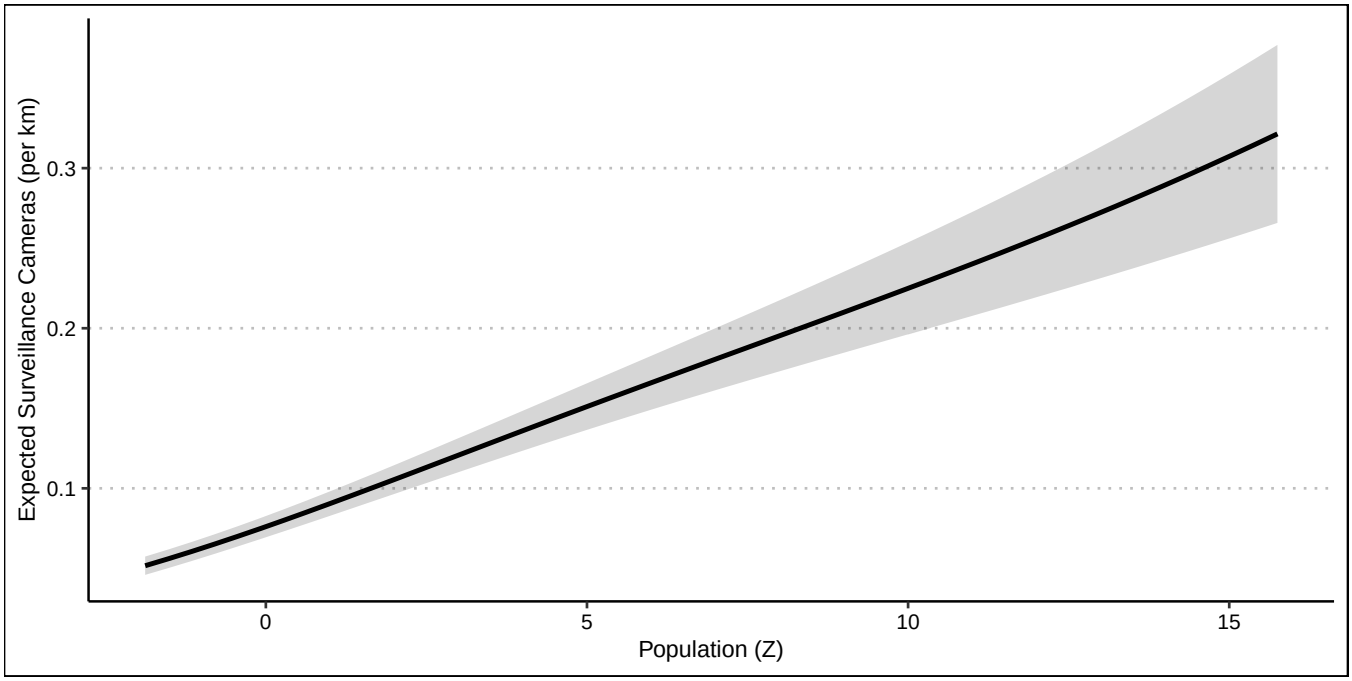


Figure 21: Population Estimated Marginal Trend (Weighted)



2.10 Median Income

Table 17: Median Income Estimated Marginal Trend (Unweighted)

Median Income (Z) Trend	SE	Z	p	Percent Change	Lower CI	Upper CI
-0.016	0.003	-4.869	<0.001	-1.54	-2.153	-0.923

Figure 22: Median Income Estimated Marginal Trend (Unweighted)

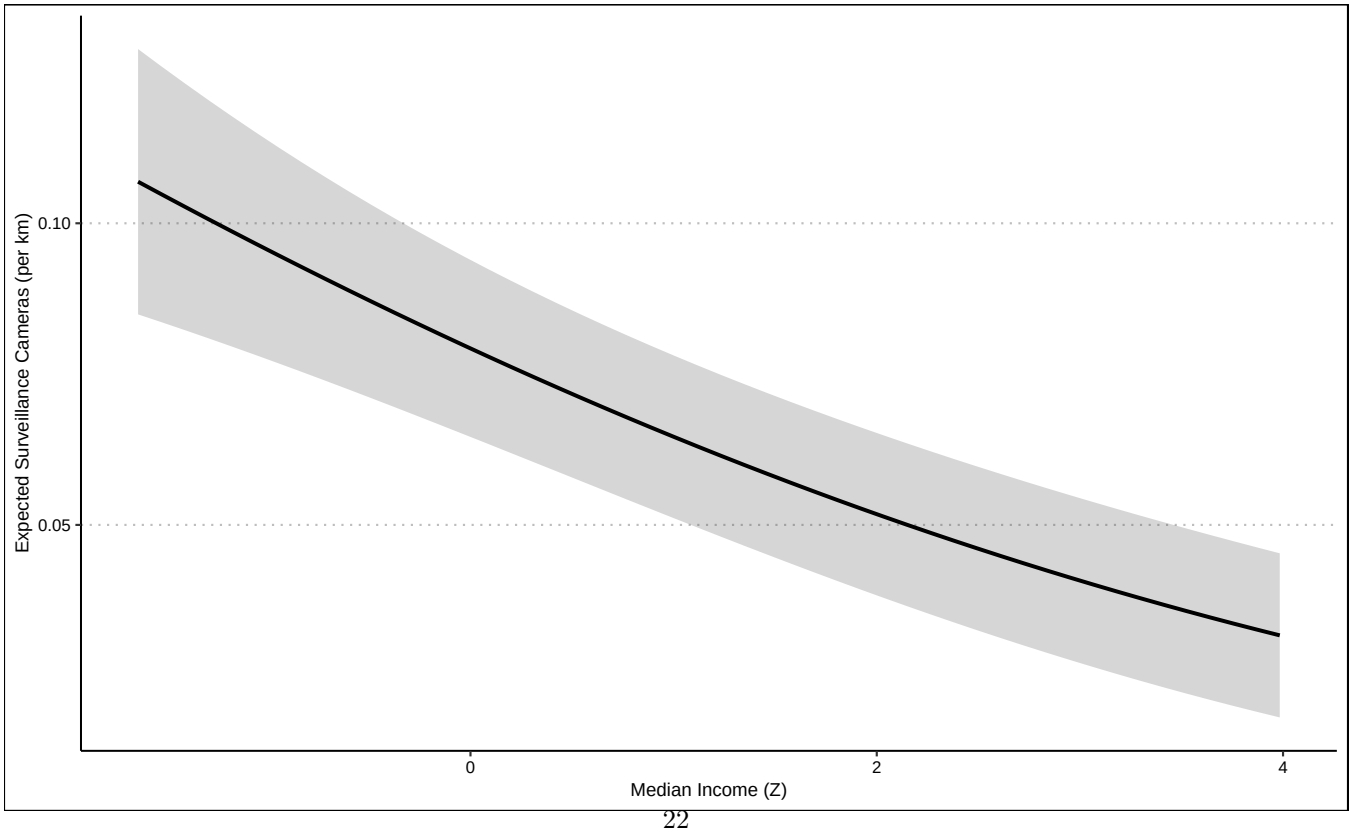
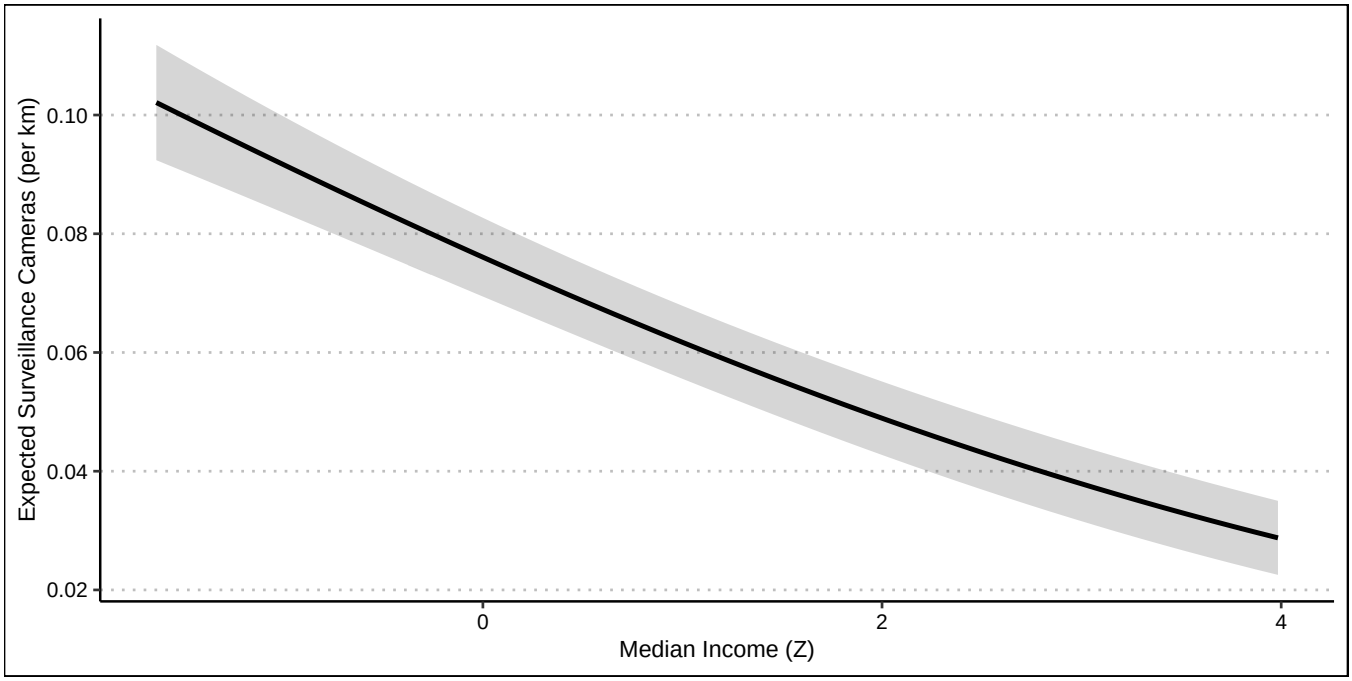


Table 18: Median Estimated Marginal Trend (Weighted)

Figure 23: Median Income Estimated Marginal Trend (Weighted)

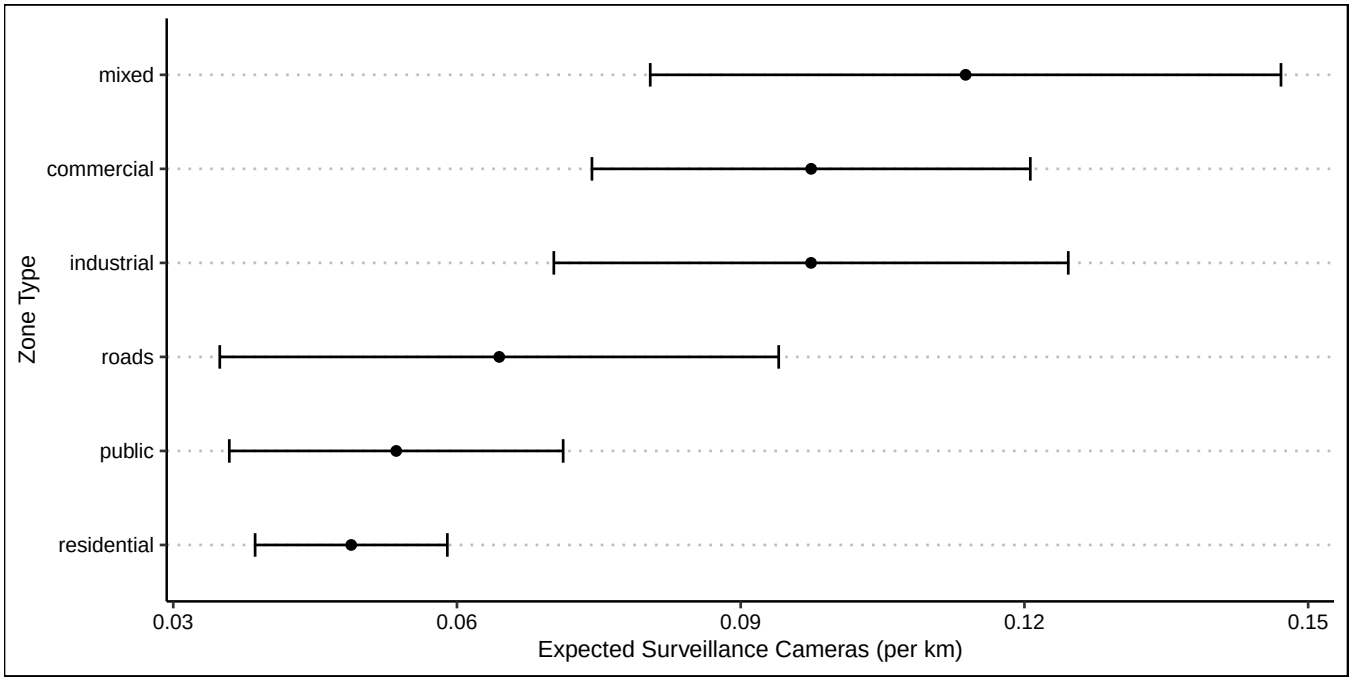


Show Contrasts Table

Table 19: Contrast of Zone Types (Unweighted)

Contrast	Estimate	SE	Z	p	Adjusted p	Percent Change	Upper CI	Lower CI
commercial - industrial	0.001	0.013	0.001	>0.999	>0.999	0.001	3.782	-3.643
commercial - mixed	0.001	0.015	-1.059	0.897	>0.999	-1.620	2.799	-5.849
commercial - public	0.044	0.011	3.838	0.002	0.006	4.485	7.944	1.136
commercial - residential	0.049	0.009	5.476	<0.001	<0.001	4.983	7.673	2.360
commercial - roads	0.033	0.020	1.687	0.54	>0.999	3.353	9.273	-2.247
industrial - mixed	0.001	0.017	-0.940	0.936	>0.999	-1.621	3.376	-6.377
industrial - public	0.044	0.014	3.129	0.022	0.054	4.484	8.741	0.393
industrial - residential	0.049	0.012	4.031	<0.001	0.004	4.982	8.653	1.435
industrial - roads	0.033	0.021	1.598	0.6	>0.999	3.352	9.609	-2.548
mixed - public	0.06	0.016	3.733	0.003	0.008	6.205	11.200	1.434
mixed - residential	0.065	0.015	4.384	<0.001	0.001	6.712	11.315	2.299
mixed - roads	0.049	0.023	2.124	0.275	0.589	5.055	12.240	-1.671
public - residential	0.005	0.008	0.619	0.99	>0.999	0.477	2.702	-1.700
public - roads	0.001	0.018	-0.610	0.99	>0.999	-1.083	4.079	-5.990
residential - roads	0.001	0.016	-0.969	0.928	>0.999	-1.553	3.082	-5.979

Figure 24: Zone Type Estimated Marginal Means (Unweighted)



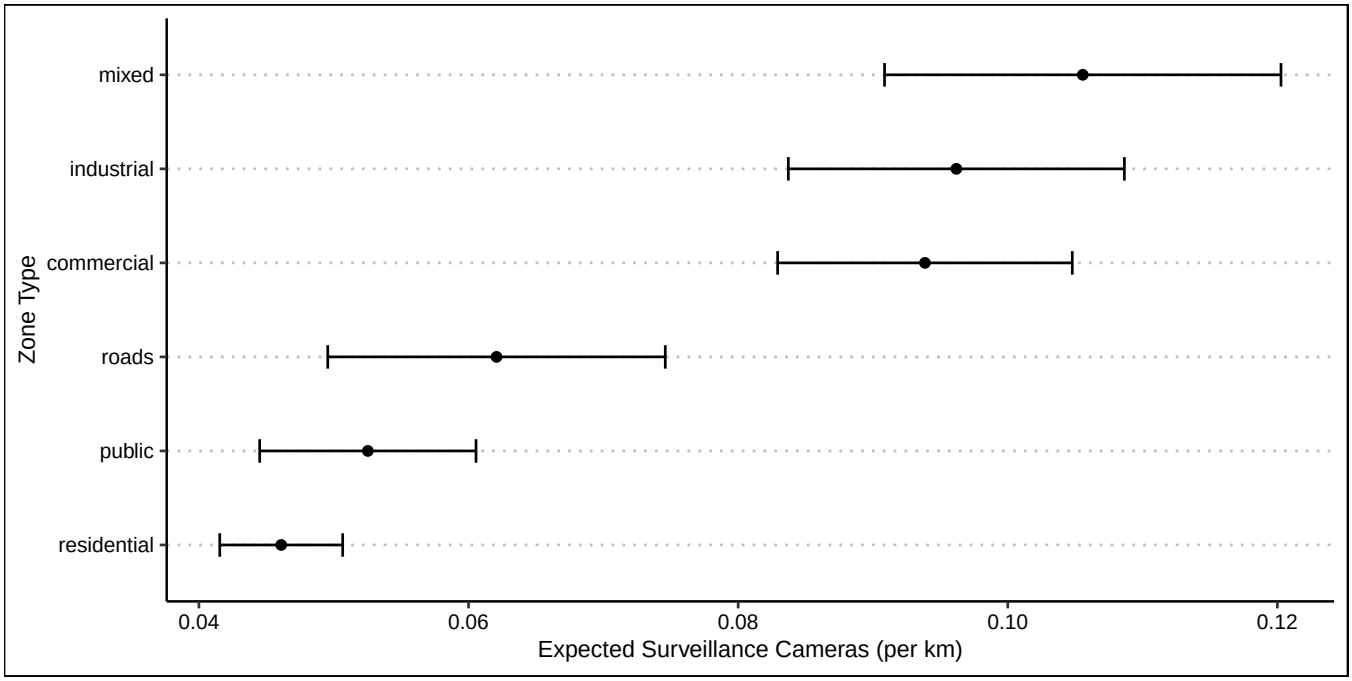
Show Contrasts table

Table 20: Contrast of Zone Types (Weighted)

Contrast	Estimate	SE	Z	p	Adjusted p	Percent Change	Upper CI	Lower CI
commercial - industrial	0.001	0.006	-0.369	>0.999	>0.999	-0.233	1.578	-2.011
commercial - mixed	0.001	0.007	-1.685	0.542	0.677	-1.164	0.813	-3.102
commercial - public	0.041	0.005	7.723	<0.001	<0.001	4.219	5.820	2.642
commercial - residential	0.048	0.004	11.184	<0.001	<0.001	4.891	6.175	3.623
commercial - roads	0.032	0.009	3.717	0.003	0.005	3.229	5.775	0.745
industrial - mixed	0.001	0.008	-1.185	0.844	0.905	-0.933	1.327	-3.143
industrial - public	0.044	0.006	6.749	<0.001	<0.001	4.462	6.406	2.554
industrial - residential	0.05	0.006	8.910	<0.001	<0.001	5.136	6.833	3.465
industrial - roads	0.034	0.009	3.799	0.002	0.004	3.470	6.152	0.856
mixed - public	0.053	0.007	7.495	<0.001	<0.001	5.446	7.594	3.341
mixed - residential	0.059	0.006	9.160	<0.001	<0.001	6.126	8.108	4.181
mixed - roads	0.043	0.010	4.372	<0.001	<0.001	4.445	7.448	1.526
public - residential	0.006	0.003	1.848	0.435	0.593	0.645	1.647	-0.348
public - roads	0.001	0.008	-1.249	0.813	0.905	-0.950	1.231	-3.084
residential - roads	0.001	0.007	-2.349	0.175	0.262	-1.584	0.341	-3.473

This is where the uncertainty differences between the models really start to be noticeable. In the weighted model, there's two clusters that are different in the number of surveilance cameras in them: Higher in the mixed, industrial, and commercial areas, and lower in the roads, public, and residential areas. However, in the unweighted model, those two groups aren't as separated, as the roads especially are harder to tell apart from the other zones.

Figure 25: Zone Type Estimated Marginal Means (Weighted)



2.12 City

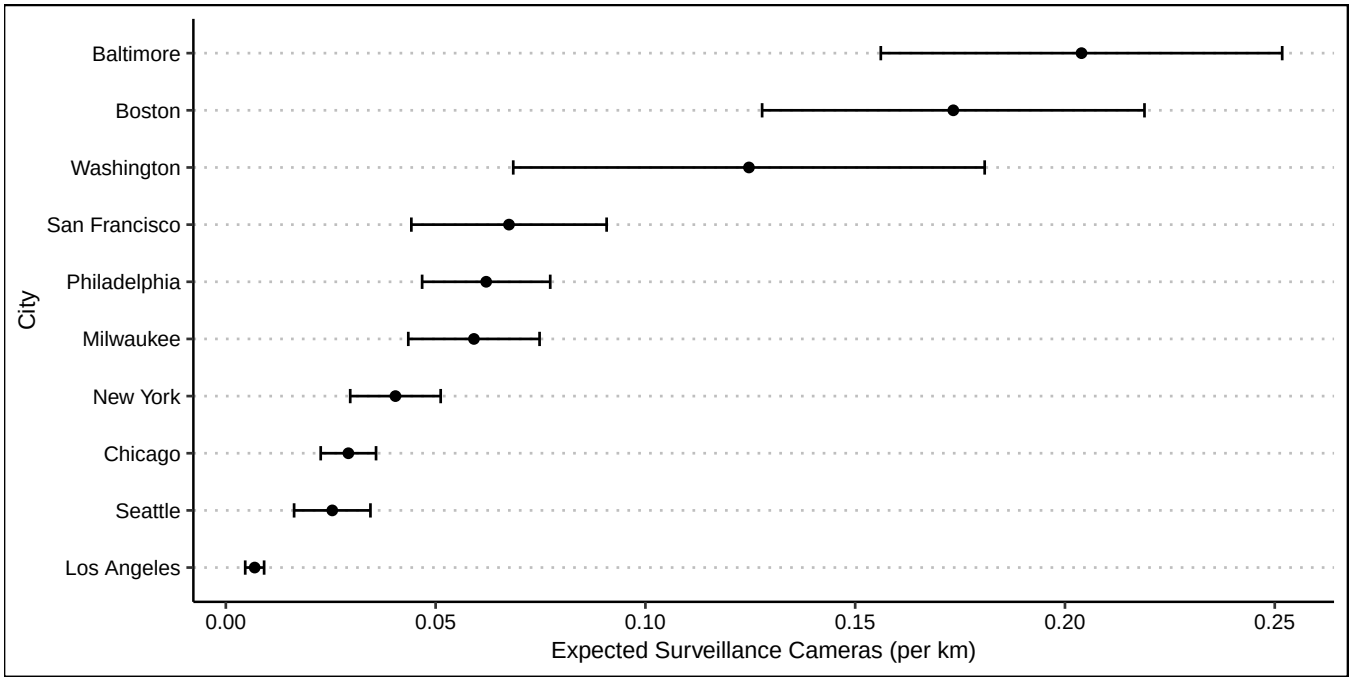
Show Contrasts table

Table 21: Contrast of Cities (Unweighted)

Contrast	Estimate	SE	Z	p	Adjusted p	Percent Change	Upper CI	Lower CI
Baltimore - Boston	0.031	0.025	1.237	0.966	>0.999	3.102	11.481	-4.647
Baltimore - Chicago	0.175	0.023	7.721	<0.001	<0.001	19.093	27.932	10.864
Baltimore - Los Angeles	0.197	0.024	8.198	<0.001	<0.001	21.782	31.405	12.864
Baltimore - Milwaukee	0.145	0.020	7.091	<0.001	<0.001	15.582	23.295	8.350
Baltimore - New York	0.164	0.022	7.302	<0.001	<0.001	17.764	26.409	9.710
Baltimore - Philadelphia	0.142	0.020	7.090	<0.001	<0.001	15.245	22.777	8.174
Baltimore - San Francisco	0.136	0.024	5.606	<0.001	<0.001	14.620	23.794	6.126
Baltimore - Seattle	0.179	0.023	7.750	<0.001	<0.001	19.550	28.590	11.146
Baltimore - Washington	0.079	0.037	2.120	0.513	0.608	8.248	21.842	-3.829
Boston - Chicago	0.144	0.022	6.628	<0.001	<0.001	15.509	23.738	7.827
Boston - Los Angeles	0.167	0.023	7.352	<0.001	<0.001	18.118	26.893	9.950
Boston - Milwaukee	0.114	0.021	5.494	<0.001	<0.001	12.104	19.728	4.965
Boston - New York	0.133	0.020	6.619	<0.001	<0.001	14.220	21.715	7.187
Boston - Philadelphia	0.111	0.020	5.521	<0.001	<0.001	11.777	19.141	4.869
Boston - San Francisco	0.106	0.020	5.330	<0.001	<0.001	11.171	18.383	4.398
Boston - Seattle	0.148	0.021	7.066	<0.001	<0.001	15.953	23.898	8.518
Boston - Washington	0.049	0.037	1.321	0.949	>0.999	4.991	17.976	-6.564
Chicago - Los Angeles	0.022	0.003	7.297	<0.001	<0.001	2.259	3.254	1.273
Chicago - Milwaukee	0.001	0.007	-4.442	<0.001	<0.001	-2.948	-0.858	-4.994
Chicago - New York	0.001	0.005	-2.408	0.321	0.412	-1.116	0.353	-2.562
Chicago - Philadelphia	0.001	0.006	-5.178	<0.001	<0.001	-3.231	-1.270	-5.153
Chicago - San Francisco	0.001	0.011	-3.432	0.021	0.034	-3.756	-0.299	-7.093
Chicago - Seattle	0.004	0.004	0.896	0.997	>0.999	0.384	1.753	-0.966
Chicago - Washington	0.001	0.029	-3.298	0.033	0.049	-9.106	-0.388	-17.061
Los Angeles - Milwaukee	0.001	0.008	-6.833	<0.001	<0.001	-5.092	-2.767	-7.360
Los Angeles - New York	0.001	0.005	-7.004	<0.001	<0.001	-3.300	-1.823	-4.754
Los Angeles - Philadelphia	0.001	0.007	-7.473	<0.001	<0.001	-5.368	-3.132	-7.553
Los Angeles - San Francisco	0.001	0.011	-5.393	<0.001	<0.001	-5.881	-2.475	-9.169
Los Angeles - Seattle	0.001	0.004	-4.458	<0.001	<0.001	-1.833	-0.536	-3.113
Los Angeles - Washington	0.001	0.029	-4.113	0.002	0.003	-11.113	-2.682	-18.814
Milwaukee - New York	0.019	0.007	2.563	0.236	0.312	1.888	4.268	-0.438
Milwaukee - Philadelphia	0.001	0.007	-0.436	>0.999	>0.999	-0.292	1.843	-2.381
Milwaukee - San Francisco	0.001	0.012	-0.674	>0.999	>0.999	-0.832	3.137	-4.649
Milwaukee - Seattle	0.034	0.008	4.478	<0.001	<0.001	3.434	5.931	0.996
Milwaukee - Washington	0.001	0.030	-2.212	0.448	0.56	-6.345	2.859	-14.725

New York - Philadelphia	0.001	0.007	-3.211	0.043	0.063	-2.139	-0.032	-4.202
New York - San Francisco	0.001	0.009	-2.914	0.101	0.138	-2.670	0.232	-5.487
New York - Seattle	0.015	0.004	3.365	0.027	0.041	1.517	2.964	0.090
New York - Washington	0.001	0.029	-2.917	0.101	0.138	-8.080	0.714	-16.107
Philadelphia - San Francisco	0.001	0.012	-0.458	>0.999	>0.999	-0.542	3.266	-4.209
Philadelphia - Seattle	0.037	0.007	5.186	<0.001	<0.001	3.736	6.083	1.441
Philadelphia - Washington	0.001	0.030	-2.123	0.511	0.608	-6.071	3.118	-14.441
San Francisco - Seattle	0.042	0.010	4.050	0.002	0.004	4.302	7.790	0.926
San Francisco - Washington	0.001	0.031	-1.867	0.692	0.798	-5.559	4.054	-14.284
Seattle - Washington	0.001	0.029	-3.438	0.021	0.034	-9.454	-0.790	-17.361

Figure 26: City Estimated Marginal Means (Unweighted)



Show Contrasts table

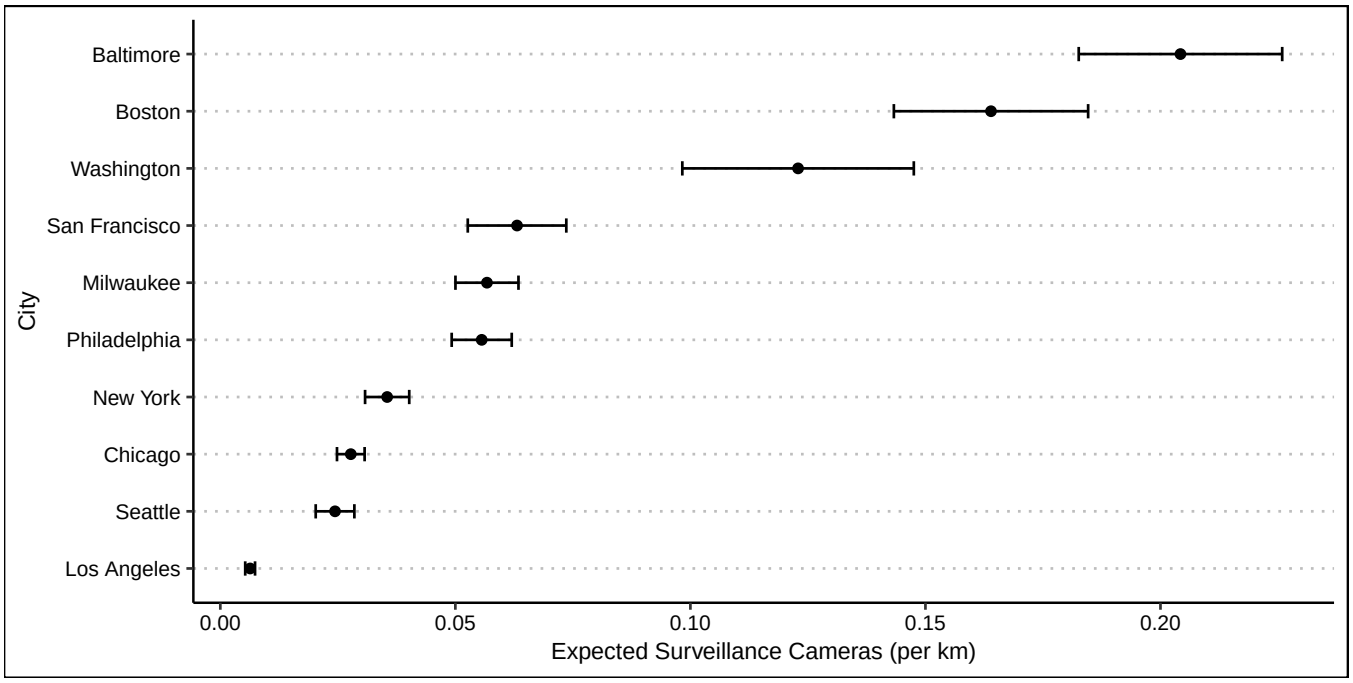
Table 22: Contrast of Cities (Weighted)

Contrast	Estimate	SE	Z	p	Adjusted p	Percent Change	Upper CI	Lower CI
Baltimore - Boston	0.04	0.011	3.652	0.01	0.011	4.113	7.812	0.541
Baltimore - Chicago	0.176	0.010	17.223	<0.001	<0.001	19.303	23.235	15.497
Baltimore - Los Angeles	0.198	0.011	18.192	<0.001	<0.001	21.886	26.154	17.762
Baltimore - Milwaukee	0.148	0.009	16.093	<0.001	<0.001	15.897	19.308	12.584
Baltimore - New York	0.169	0.010	16.592	<0.001	<0.001	18.384	22.256	14.635

Baltimore - Philadelphia	0.149	0.009	16.279	<0.001	<0.001	16.027	19.427	12.723
Baltimore - San Francisco	0.141	0.011	12.789	<0.001	<0.001	15.158	19.249	11.207
Baltimore - Seattle	0.18	0.010	17.343	<0.001	<0.001	19.704	23.697	15.841
Baltimore - Washington	0.081	0.016	4.950	<0.001	<0.001	8.475	14.264	2.980
Boston - Chicago	0.136	0.010	13.746	<0.001	<0.001	14.590	18.239	11.054
Boston - Los Angeles	0.158	0.010	15.355	<0.001	<0.001	17.071	20.935	13.330
Boston - Milwaukee	0.107	0.009	11.395	<0.001	<0.001	11.319	14.682	8.054
Boston - New York	0.128	0.009	13.991	<0.001	<0.001	13.707	17.059	10.452
Boston - Philadelphia	0.108	0.009	11.727	<0.001	<0.001	11.443	14.749	8.233
Boston - San Francisco	0.101	0.009	11.271	<0.001	<0.001	10.609	13.784	7.522
Boston - Seattle	0.14	0.009	14.870	<0.001	<0.001	14.975	18.440	11.612
Boston - Washington	0.041	0.016	2.545	0.244	0.268	4.190	9.642	-0.992
Chicago - Los Angeles	0.021	0.001	15.495	<0.001	<0.001	2.165	2.612	1.719
Chicago - Milwaukee	0.001	0.003	-10.095	<0.001	<0.001	-2.855	-1.969	-3.733
Chicago - New York	0.001	0.002	-3.679	0.009	0.01	-0.770	-0.108	-1.428
Chicago - Philadelphia	0.001	0.003	-10.425	<0.001	<0.001	-2.746	-1.921	-3.565
Chicago - San Francisco	0.001	0.005	-6.994	<0.001	<0.001	-3.475	-1.918	-5.007
Chicago - Seattle	0.003	0.002	1.736	0.775	0.83	0.336	0.951	-0.275
Chicago - Washington	0.001	0.013	-7.516	<0.001	<0.001	-9.076	-5.361	-12.646
Los Angeles - Milwaukee	0.001	0.003	-15.375	<0.001	<0.001	-4.914	-3.923	-5.894
Los Angeles - New York	0.001	0.002	-13.888	<0.001	<0.001	-2.873	-2.226	-3.516
Los Angeles - Philadelphia	0.001	0.003	-15.941	<0.001	<0.001	-4.807	-3.872	-5.733
Los Angeles - San Francisco	0.001	0.005	-11.249	<0.001	<0.001	-5.520	-3.999	-7.017
Los Angeles - Seattle	0.001	0.002	-9.683	<0.001	<0.001	-1.790	-1.209	-2.368
Los Angeles - Washington	0.001	0.013	-9.296	<0.001	<0.001	-11.003	-7.401	-14.464
Milwaukee - New York	0.021	0.003	6.742	<0.001	<0.001	2.146	3.169	1.133
Milwaukee - Philadelphia	0.001	0.003	0.404	>0.999	>0.999	0.112	0.994	-0.762
Milwaukee - San Francisco	0.001	0.006	-1.158	0.978	>0.999	-0.638	1.115	-2.360
Milwaukee - Seattle	0.032	0.003	10.071	<0.001	<0.001	3.285	4.339	2.242
Milwaukee - Washington	0.001	0.013	-5.130	<0.001	<0.001	-6.404	-2.505	-10.147
New York - Philadelphia	0.001	0.003	-6.921	<0.001	<0.001	-1.991	-1.086	-2.888
New York - San Francisco	0.001	0.004	-6.455	<0.001	<0.001	-2.725	-1.399	-4.034
New York - Seattle	0.011	0.002	5.618	<0.001	<0.001	1.115	1.748	0.486
New York - Washington	0.001	0.013	-6.943	<0.001	<0.001	-8.370	-4.647	-11.949
Philadelphia - San Francisco	0.001	0.005	-1.414	0.923	0.966	-0.749	0.934	-2.404
Philadelphia - Seattle	0.031	0.003	10.498	<0.001	<0.001	3.169	4.144	2.204
Philadelphia - Washington	0.001	0.013	-5.250	<0.001	<0.001	-6.509	-2.639	-10.225
San Francisco - Seattle	0.039	0.005	8.378	<0.001	<0.001	3.948	5.479	2.439
San Francisco - Washington	0.001	0.013	-4.486	<0.001	<0.001	-5.803	-1.747	-9.692

Seattle - Washington	0.001	0.013	-7.816	<0.001	<0.001	-9.381	-5.695	-12.923

Figure 27: Cities Estimated Marginal Means (Weighted)



Differences in individual cities are much less obvious in the unweighted model. In both models, Baltimore, Boston, and Washington D.C. are on the higher end of the scale for surveillance cameras. San Francisco, Milwaukee, and Philadelphia are all very similar to each other. Lastly, Los Angeles clearly has the least amount of the surveillance cameras.

2.13 Conclusions

We have strong evidence that the more heavily an area is made up of black residents, there are fewer surveillance cameras (Figure 14; Figure 15). If we could connect this with strong evidence that racial diversity affects the prevalence of surveillance cameras, we could start getting at the researcher's hypothesis, but I hesitate to make that claim. Because of the differences in the weighted and unweighted model, we see that those observations that are weighted higher are entirely driving the effect of diversity (Figure 16; Figure 17).

Somewhat expectedly, areas that have high rates of crime also have high numbers of surveillance cameras (Figure 18; Figure 19), though there's a very high range of uncertainty as crime rates become higher.

Going along with crime rate, higher population areas have higher numbers of surveillance cameras (Figure 18; Figure 19), most likely for the reasons that more people means more crime and other things to surveill.

Inversely to crime rate, wealthier areas tend to have lower numbers of surveillance cameras (Figure 22; Figure 23), probably for similar reasons.

The zone of the road makes a difference to the number of surveillance cameras (Figure 24; Figure 25). Even though we're hearing about the increasing prevalence of surveillance cameras in residential neighborhoods, the data still suggests that those zones have the fewest amount of cameras.

Lastly, individual cities have quite large differences in the number of surveillance cameras (Figure 26; Figure 27). This shouldn't be surprising. As we saw in the VIF table (Table 1), the cities are highly correlated with the other factors, so we can almost view each city as just the products of demographic and socioeconomic features that could be hard to separate from individual political or geographic effects.

We should probably talk about the weights. Usually, in the case of a meta-analysis, the sample size of each study being the weights directly corresponds to reliability in a statistical sense, but that's just not the case in this study.

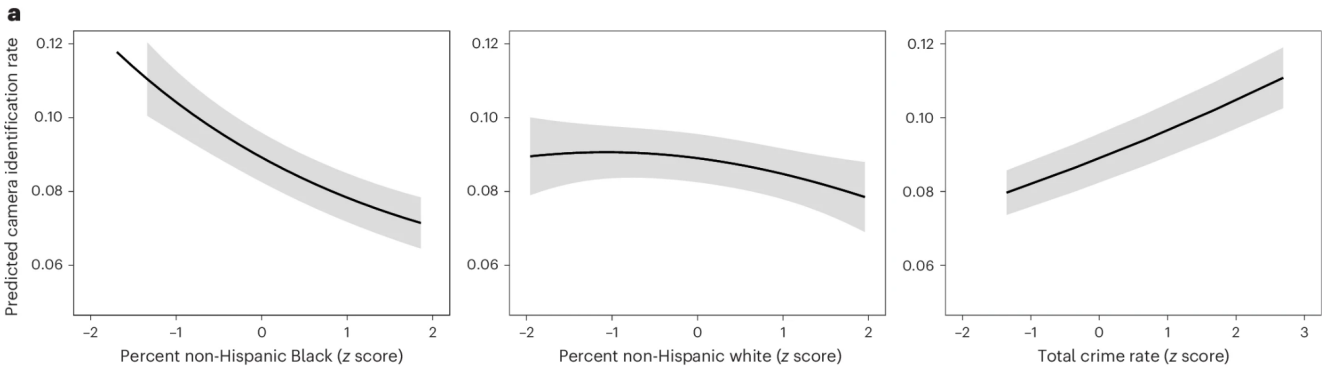
The randomly sampled nature of the street view images mean there is no way to know if 10 images are twice as reliable as 5. Based on where those images were taken, they could give the exact same amount of information, but not the same power over the model.

My gut tells me to go with the results that hold between the two models, so that includes the effects of Black % (Figure 14; Figure 15), crime rate (at high levels) (Figure 18; Figure 19), population (Figure 20; Figure 21), median income (Figure 22; Figure 23), zoning (Figure 24; Figure 25), and cities (Figure 26; Figure 27).

3 Comparisons With the Article

The main figure we can compare our results to in the paper is Figure 2, which I'll split up by section. A difference that may or may not be visible is the fact that they authors actually have two separate models models, one with diversity, and one with individual black % white % variables.

Figure 28: Dahir et al.



The models shown in Figure 2a only use diversity, and not black or white % variables.

Compared to our model, which includes the individual black and white % terms and diversity, their figure shows more of a quadratic shape, whereas ours is more linear appearing (Figure 14; Figure 15). Still, our conclusion is the same.

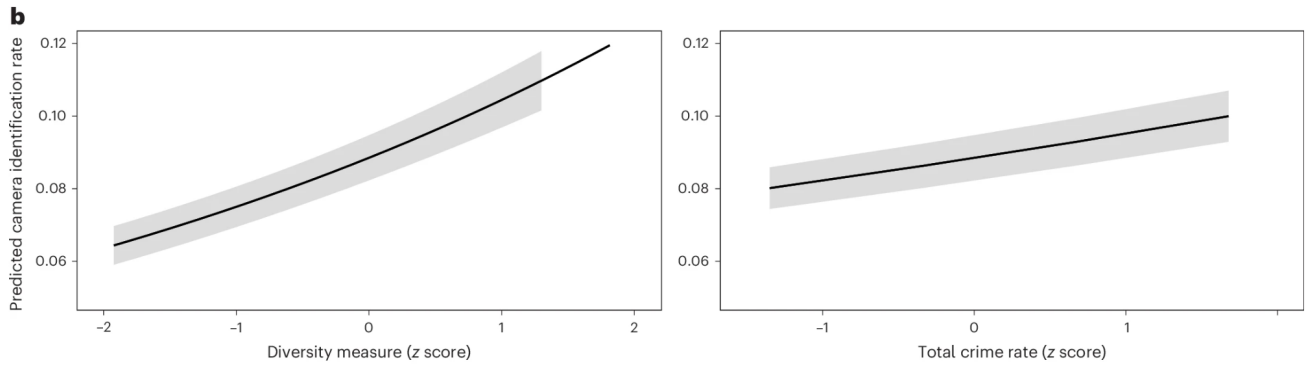
Because the only indication of the white % having an effect was a marginal effect in our weighted model (Table 8), I didn't see a need to further analyze it. So while the authors claim:

> "Fig. 2a (middle) reveals a nonlinear relationship between the percentage of non-Hispanic white residents and the estimated camera identification rate: the camera identification rate is constant with the percentage of non-Hispanic white residents up to a certain point (around the city mean) after which it starts to decline, although estimates are noisy. Thus, neighborhoods with low-to-moderate proportions of white residents appear to have higher camera detection rates than those with very high proportions of white residents."

Because they don't give any statistical evidence for this claim, like a discernible effect of the coefficient, I wonder if they're just interpreting the shape of the graph. Even then, the graph seems relatively flat which I would interpret as showing little to no effect.

Lastly, for the crime rate graph, they severly truncate their graph to only show a maximum of 2.5 standard deviations above the mean, so there's no way to see the, in my opinion, more interesting relationship between crime and surveillance cameras in especially high crime rate areas. Even if the tail of our graph near 40 standard deviations is due to one outlier (Figure 18; Figure 19), there are still a good amount of observations over 10 standard deviations above the mean, so it's a shame they don't show that. Still, the results are pretty similar between our analyses.

Figure 29: Dahir et al.

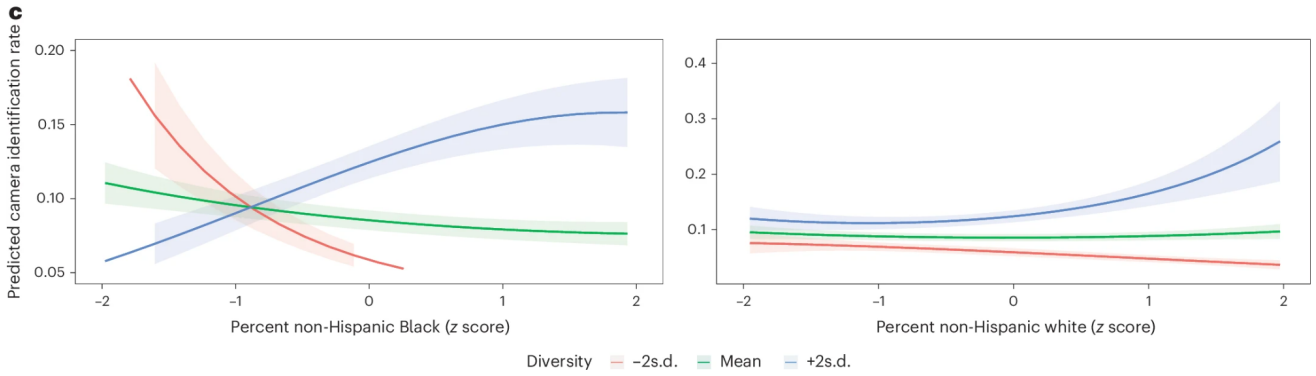


In Figure 2b they show their model that replaces the black % white % variables with an overall shannon diversity metric. In our analysis, only the weighted model showed a discernible effect of diversity (Table 8), though that was with all the terms included. Due to the high colinearity between the black & white % variables and diversity (Table 1), it's reasonable to assume our model was unable to separate those. So, while our plot looks similar to there's, I wouldn't say the evidence across both the unweighted and weighted model is strong. So when the authors say:

> "As shown, camera detection is monotonically positively related to neighborhood diversity, indicating that more diverse neighborhoods tend to have higher rates of camera detection."

I'd have to push back on that slightly, as that is really heavily dependent on the model weights.

Figure 30: Dahir et al.



Even though the authors don't add any interaction terms to their model, or discuss that in the methods, the seem to be testing the interaction between black % and diversity and the interaction between white % and diversity. To compare, I'll attempt to recreate their analysis:

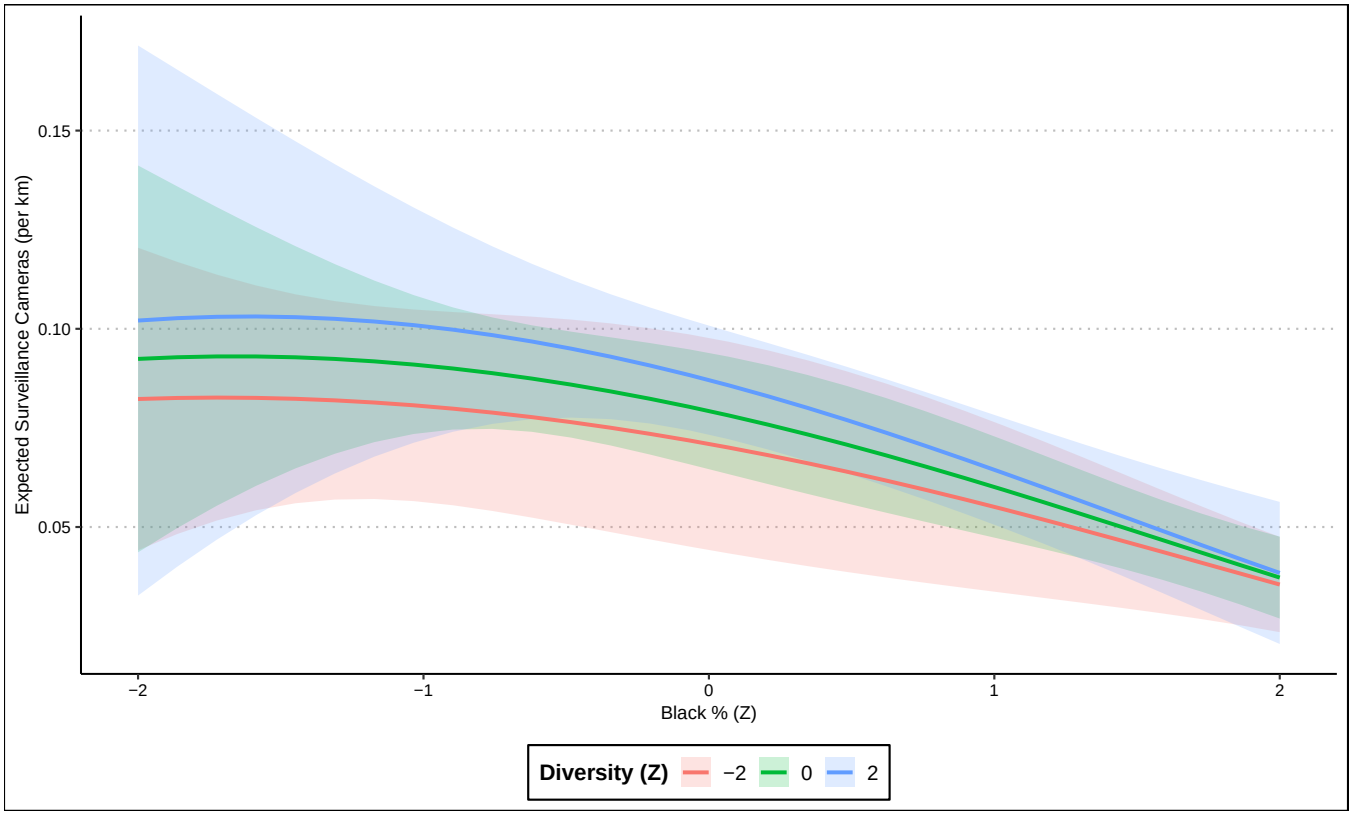
For the unweighted model:

Table 23: Contrasts of Estimated Marginal Trends of Black % (Z) Across Diversity (Unweighted; Post-hoc)

Contrast	Estimate	SE	Z	p	adjusted-p	Percent Change	Lower CI	Upper CI
Low (-2SD) - Average	0.003	0.002	1.372	0.355	0.448	0.279	-2.560	0.937
Low (-2SD) - High (2SD)	0.006	0.005	1.290	0.401	0.448	0.583	-5.420	2.378
Average - High (2SD)	0.003	0.003	1.208	0.448	0.448	0.303	-2.989	1.484

To recreate their plot:

Figure 31: Black % Estimated Marginal Trends Across Diversity (Unweighted)

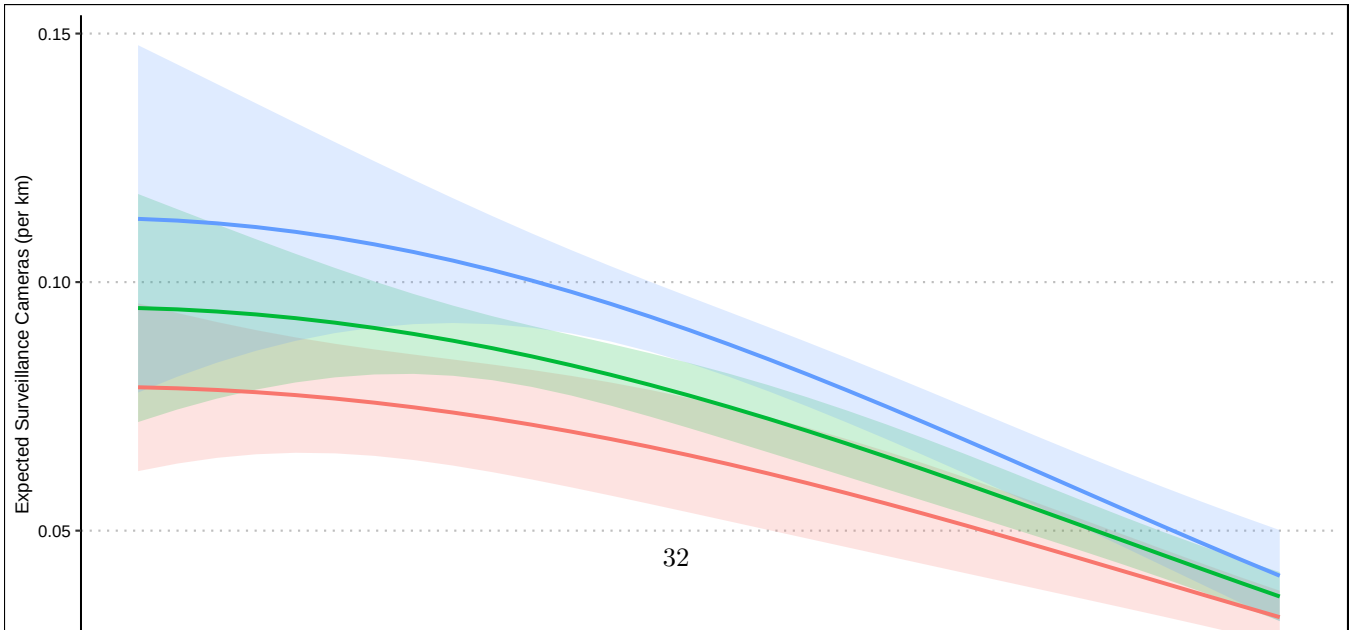


Now for the weighted model:

Table 24: Contrasts of Estimated Marginal Trends of Black % (Z) Across Diversity (Weighted; Post-hoc)

Contrast	Estimate	SE	Z	p	adjusted-p	Percent Change	Lower CI	Upper CI
Low (-2SD) - Average	0.004	0.001	3.729	0.001	0.002	0.355	-1.881	-0.455
Low (-2SD) - High (2SD)	0.008	0.002	3.448	0.002	0.002	0.779	-4.159	-0.681
Average - High (2SD)	0.004	0.001	3.217	0.004	0.004	0.423	-2.339	-0.210

Figure 32: Black % Estimated Marginal Trends Across Diversity (Weighted)



project possible.

5 AI Use

All of the guidance given by generative AI for this article can be found in this conversation.

6 Statistical Packages Used

- *tidyverse* for... everything it's *tidyverse* (Wickham et al. 2019)
- *ggplot2* for plotting (Wickham 2016)
- *ggthemes* for *theme_clean()* (Arnold 2024)
- *patchwork* for combined plots (Pedersen 2025)
- *pscl* for inference of Zero Inflated Count Regression model (Zeileis et al. 2008)
- *performance* for model fit statistics (Lüdecke et al. 2021)
- *emmeans* for estimated marginal means and trends (Lenth 2025)

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